Locating Bragg Disks in CBED Patterns with a Disk–Ring Filter, and Why Disk Overlap Makes It Hard

Diffraction project

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Abstract

Bragg disks in a convergent-beam electron diffraction (CBED) pattern can be located without training by two steps: a logarithmic intensity scale with a floor relative to the brightest pixel (*true log*), and correlation with a zero-mean template that compares a disk of the probe radius with a thin ring around it (*disk–ring filter*). We describe the method and evaluate it on 1 875 simulated patterns (25 crystals, three probe angles, 25 thicknesses), scoring every pattern separately. The single quantity that decides whether it works is the *overlap ratio* $\eta = 2r/s$, the disk diameter divided by the spacing between neighbouring reflections. While $\eta \lesssim 0.8$ the filter's ring lies on background and almost every visible reflection is found, with a localisation error of 0.2 px to 0.3 px. Once the ring reaches the neighbouring disks, their intensity is subtracted from the response at the true centre: the response there falls from a robust z-score of about 5 to near 0, the error grows to 1 px, and fewer than 5 % of the lattice reflections are found when $\eta > 1.5$. A synthetic experiment that changes only the disk size reproduces the collapse, and shows that interference between overlapping disks makes it worse. Matching the probe rim with Sobel edges instead doubles the centres found at 4 mrad but halves them at 2 mrad. To keep the disks separable, the probe semi-angle should not exceed $(s - 3\,\mathrm{px})/(2\kappa)$, where $\kappa \approx 2.5\,\mathrm{px/mrad}$ is the disk radius per milliradian.

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1 Introduction

In CBED the incident beam is a cone of semi-angle α, so every Bragg reflection appears as a disk rather than a spot. The disk radius is set by the probe and grows with α; the distance between disks is set by the crystal lattice and does not. Locating the disks therefore becomes harder as α grows, because neighbouring disks approach, touch and finally overlap (Figure 1).

This tutorial has two aims. The first is to describe a simple, training-free detector, true-log scaling followed by a disk–ring matched filter (Section 3), and to show how well it works when the disks are separate (Section 5). The second, and main, aim is to explain *why* it fails when the disks overlap (Section 6), and to test whether matching only the disk rims helps (Section 7). Everything is organised around one number, the overlap ratio

$$\eta = \frac{2r}{s}, \tag{1}$$

where r is the disk radius and s the distance between neighbouring reflections, both in pixels. Neighbouring disks touch at $\eta = 1$. Section 8 turns the analysis into a rule for choosing α.

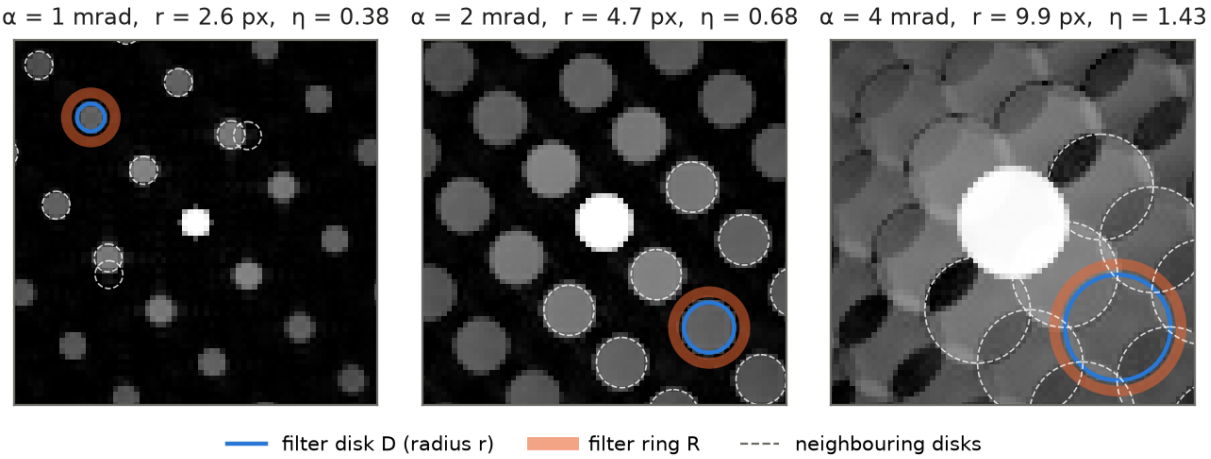

Figure 1: The same crystal ($ZrSn_3$, 20 Å) at three probe semi-angles. The lattice is unchanged; only the disks grow. The filter of Section 3 compares the mean inside a disk (blue) with the mean in a thin ring around it (orange). At 1 and 2 mrad the ring lies on dark background; at 4 mrad ($\eta = 1.43$) it lies on the neighbouring disks (dashed).

2 Data and notation

The patterns are multislice simulations (pyprismatic) of SiO_2 and 24 further crystals, each at probe semi-angles $\alpha = 1$, 2 and 4 mrad and 25 thicknesses $t = 20, 40, \ldots, 500$ Å: 1 875 patterns in total. Each simulation provides three arrays: M, the diffraction pattern; V, a map whose local maxima mark the reciprocal-lattice points, used only for the ground truth; and P, the probe aperture, of which every disk is a shifted copy. All processing uses the central 128×128 pixels of the 256×256 detector ($0.02\,\text{Å}^{-1}$ per pixel). Table 1 lists the symbols used throughout.

Table 1: Notation.

Symbol	Meaning
M, V, P	diffraction pattern, lattice-point map (ground truth only), probe aperture
T	true-log image, Equation (2)
α, t	probe semi-angle (mrad), sample thickness (Å)
r, κ	disk radius (px), from the probe, Equation (3); $r \approx \kappa\alpha$
s	nearest-neighbour distance between lattice points (px)
$\eta = 2r/s$	overlap ratio; disks touch at $\eta = 1$
D, R; g, w	filter disk and ring; gap and ring width ($g = 1$ px, $w = 2$ px)
K, S, z	kernel, filter response, robust z-score of the response
R_{vis}, R_{all}	per-image recall of visible reflections and of all lattice reflections, Section 4

3 Method: true log and disk–ring filter

3.1 True-log intensity scaling

Disk intensities span several orders of magnitude: about 1 for the direct beam, 10^{-2} for strong reflections and 10^{-4} or less for weak ones (Figure 3a). The common transform $\log(1+x)$ does not help here, because $\log(1+x) \approx x$ for $x \ll 1$. Instead the logarithm is taken above a floor proportional to the brightest pixel. With $x_{ij} = \max(M_{ij}, 0)$ and $m = \max_{ij} x_{ij}$,

$$T_{ij} = \text{clip}\left(\frac{\log_{10}(x_{ij} + \varepsilon m) - \log_{10}(\varepsilon m)}{\log_{10}(m + \varepsilon m) - \log_{10}(\varepsilon m)},\ 0,\ 1\right), \qquad \varepsilon = 10^{-4}. \tag{2}$$

This maps the four decades between εm and m onto $[0, 1]$, so weak and strong disks become comparable (Figure 3b). A larger ε hides weak disks; a smaller one lifts the diffuse background above the floor.

3.2 Disk radius from the probe

The aperture image is close to binary. With $w(\boldsymbol{u}) = \mathbf{1}[P(\boldsymbol{u}) > 0.5]$ the disk radius follows from its area,

$$r = \sqrt{\frac{1}{\pi} \sum_{\boldsymbol{u}} w(\boldsymbol{u})}, \tag{3}$$

giving $r = 2.6$, 4.7 and 9.9 px at $\alpha = 1$, 2 and 4 mrad, that is $r \approx \kappa\alpha$ with $\kappa \approx 2.5\,\text{px/mrad}$. The radius is the filter's only size parameter.

3.3 Disk–ring matched filter

A disk is a bright region of radius r on a darker background. The filter compares a disk D with a ring R outside it, separated by a gap g so that the soft disk edge falls in neither:

$$D = \{\boldsymbol{u} : \|\boldsymbol{u}\| \le r\}, \qquad R = \{\boldsymbol{u} : r + g < \|\boldsymbol{u}\| \le r + g + w\}. \tag{4}$$

The kernel is positive on D, negative on R, and sums to zero (Figure 2):

$$K(\boldsymbol{u}) = \frac{\mathbf{1}_D(\boldsymbol{u})}{|D|} - \frac{\mathbf{1}_R(\boldsymbol{u})}{|R|}. \tag{5}$$

The response at pixel $\boldsymbol{p}$ is therefore the mean of T over the disk minus its mean over the ring:

$$S(\boldsymbol{p}) = \sum_{\boldsymbol{u}} K(\boldsymbol{u})\, T(\boldsymbol{p} + \boldsymbol{u}) = \underbrace{\frac{1}{|D|} \sum_{\boldsymbol{u} \in D} T(\boldsymbol{p} + \boldsymbol{u})}_{\text{disk mean}} - \underbrace{\frac{1}{|R|} \sum_{\boldsymbol{u} \in R} T(\boldsymbol{p} + \boldsymbol{u})}_{\text{ring mean}}. \tag{6}$$

S is computed for all pixels at once by FFT convolution. The image is first extended by mirror reflection over the kernel half-width, so that near the border the ring samples image-like values instead of zeros.

Two properties make this a matched filter for a disk. A uniform region gives $S = 0$, so smooth background is ignored; and an isolated disk of height h above its surroundings gives $S = h$ exactly at its centre, falling off-centre as part of the disk enters the ring. Each disk therefore produces one maximum at its centre (Figure 3c). Both properties rely on one assumption: *the ring lies on background*. Section 6 is about what happens when it does not.

3.4 From response to disk centres

The response is converted to a robust z-score, so that one threshold scale applies to all images:

$$z(\boldsymbol{p}) = \frac{S(\boldsymbol{p}) - \operatorname{med}(S)}{1.4826\, \operatorname{med}(|S - \operatorname{med}(S)|)}. \tag{7}$$

Candidate centres are the local maxima of z in a 9×9 window (minimum separation $\rho = 4\,\text{px}$), refined to sub-pixel precision by fitting a parabola through each maximum and its two neighbours along each axis. Each candidate keeps its z value as confidence; a threshold on z selects the reported disks.

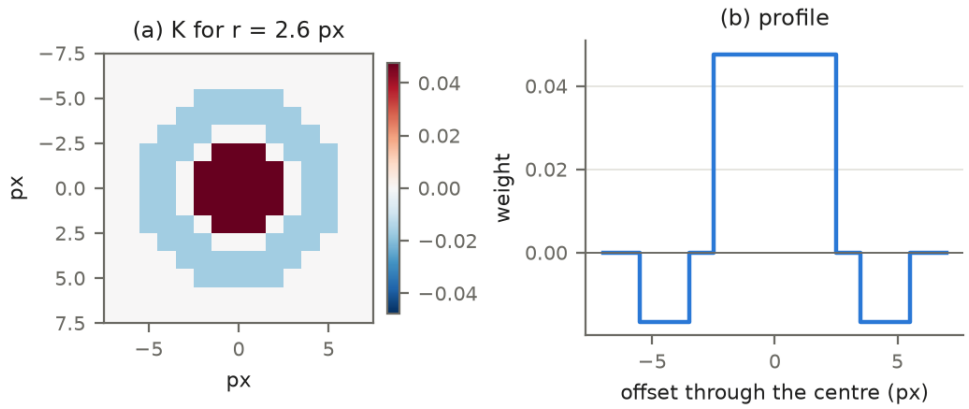

Figure 2: Disk–ring kernel K for $r = 2.6$ px (1 mrad): (a) weights, (b) profile through the centre. Positive on the disk, negative on the ring, zero in the gap; the weights sum to zero.

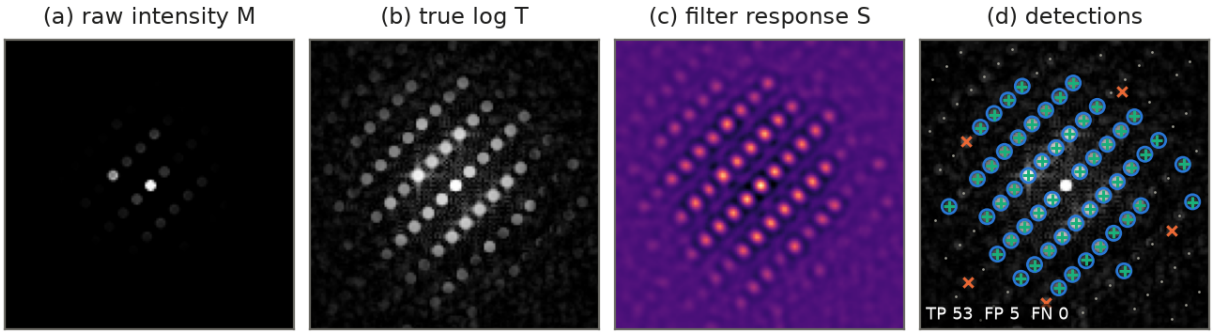

Figure 3: The method on one pattern (SiO_2, 1 mrad, 200 Å): (a) raw intensity, dominated by the direct beam; (b) true log, Equation (2); (c) filter response, Equation (6), one peak per disk; (d) detections at the threshold of this image (symbols as in Figure 4).

4 Evaluation

Lattice reflections. The ground-truth centres are the local maxima of V (in a 5×5 window, at least 5 % of the strongest reflection, direct beam excluded), refined by a 3×3 centroid. They do not depend on α or t and mark every reflection of the lattice, whether or not it is bright in M. The spacing s is the median distance from each lattice point to its nearest neighbour.

Visible reflections. Many lattice reflections carry almost no intensity in M. A reflection is called *visible* if, in M, the mean over its disk is at least three times the mean over its ring (Equation (4)) and at least 10^{-3} of the brightest disk.

Matching. Detections are taken in order of decreasing z; one is correct if an unmatched reflection lies within 2 px. A detection on the direct beam is ignored.

Per-image recall. Every image (one crystal, angle and thickness) is scored on its own; nothing is pooled across thicknesses or angles. For each image the threshold is the lowest z that keeps at most five false detections in that image, and two recalls are reported:

$$R_{\mathrm{vis}} = \frac{\text{visible reflections found}}{\text{visible reflections}}, \qquad R_{\mathrm{all}} = \frac{\text{lattice reflections found}}{\text{lattice reflections}}. \tag{8}$$

R_{vis} is computed only for images with at least three visible reflections. Summaries over crystals or thicknesses are medians of these per-image values. The two recalls answer different questions: R_{vis} asks how well the detector finds what is plainly there, R_{all} how much of the lattice it recovers. Section 6.5 shows why only R_{all} reveals the effect of overlap.

5 Results

5.1 Three example crystals

Table 2 follows three crystals across the three probe angles. SiO_2 has a dense lattice ($s = 8.6$ px); $AlCo_3$ and $ZrSn_3$ have wider spacings ($s \approx 14$ px). Whenever $\eta < 0.8$, the filter response at the true centres is high ($z = 2.4$–5.8), almost every visible reflection is found ($R_{\mathrm{vis}} = 0.94$–1.00) and the correct detections lie within a quarter of a pixel at 1 mrad. When $\eta > 1$ the picture changes in every column at once: the ring lies on neighbouring disks, the response at the centres drops to 1.2 or less, few reflections remain visible, and R_{all} falls to 0.02–0.30. Figure 4 shows SiO_2 image by image.

5.2 All 24 crystals

Figure 5 shows R_{vis} for every image of the 24 crystals, with the crystals ordered by their spacing s and the overlap ratio printed next to each row. Two patterns stand out. Along each row, recall varies with thickness but rarely collapses. Down the columns, the number of images that can be scored at all falls with η: at 4 mrad, where every crystal has $\eta > 1$, only 259 of 600 images contain three or more visible reflections, and the crystals with the densest lattices have almost none. Table 3 and Figure 6 give the thickness dependence: R_{vis} decreases slowly with thickness at every angle, while R_{all} is set mainly by the angle (median 0.47, 0.25 and 0.04 at 200 Å for 1, 2 and 4 mrad).

Table 2: Three crystals at three probe angles. Values are medians over the 25 thicknesses of per-image quantities. Ring on neighbours: fraction of the filter ring lying inside other disks. z at centres: filter response at the true lattice points, Equation (7). Error: distance of correct detections to the true centre (1 px = 0.02 Å^{-1}). –: fewer than three visible reflections in (almost) every image.

Crystal	α (mrad)	s (px)	η	Ring on neighb.	z at centres	Visible per image	R_{vis}	R_{all}	Error (px)
SiO_2	1	8.6	0.60	0.00	4.13	63	0.97	0.65	0.20
	2		1.09	0.36	1.23	23	0.96	0.30	0.51
	4		2.29	1.00	0.26	0	–	0.03	–
$AlCo_3$	1	14.0	0.37	0.00	4.35	58	0.94	0.75	0.16
	2		0.67	0.00	2.40	43	0.94	0.54	0.49
	4		1.41	0.99	0.13	5	0.88	0.08	0.38
$ZrSn_3$	1	13.8	0.38	0.00	5.76	59	1.00	0.94	0.22
	2		0.68	0.00	2.46	50	0.97	0.72	0.42
	4		1.43	1.00	0.25	0	–	0.02	–

Table 3: 24 crystals, per thickness: median over the crystals of the per-image recalls. R_{vis} uses images with at least three visible reflections (number of crystals in brackets); R_{all} uses all images. Median η: 0.62, 1.13 and 2.37 at 1, 2 and 4 mrad.

	R_{vis}			R_{all}		
t (Å)	1 mrad	2 mrad	4 mrad	1 mrad	2 mrad	4 mrad
20	0.93 (24)	1.00 (22)	1.00 (11)	0.74	0.33	0.08
100	0.93 (24)	0.93 (23)	0.86 (12)	0.57	0.34	0.05
200	0.91 (24)	0.91 (24)	0.83 (10)	0.47	0.25	0.04
300	0.86 (24)	0.78 (23)	0.68 (10)	0.34	0.16	0.02
400	0.87 (24)	0.80 (23)	0.67 (11)	0.26	0.13	0.02
500	0.88 (24)	0.81 (23)	0.72 (10)	0.27	0.13	0.02

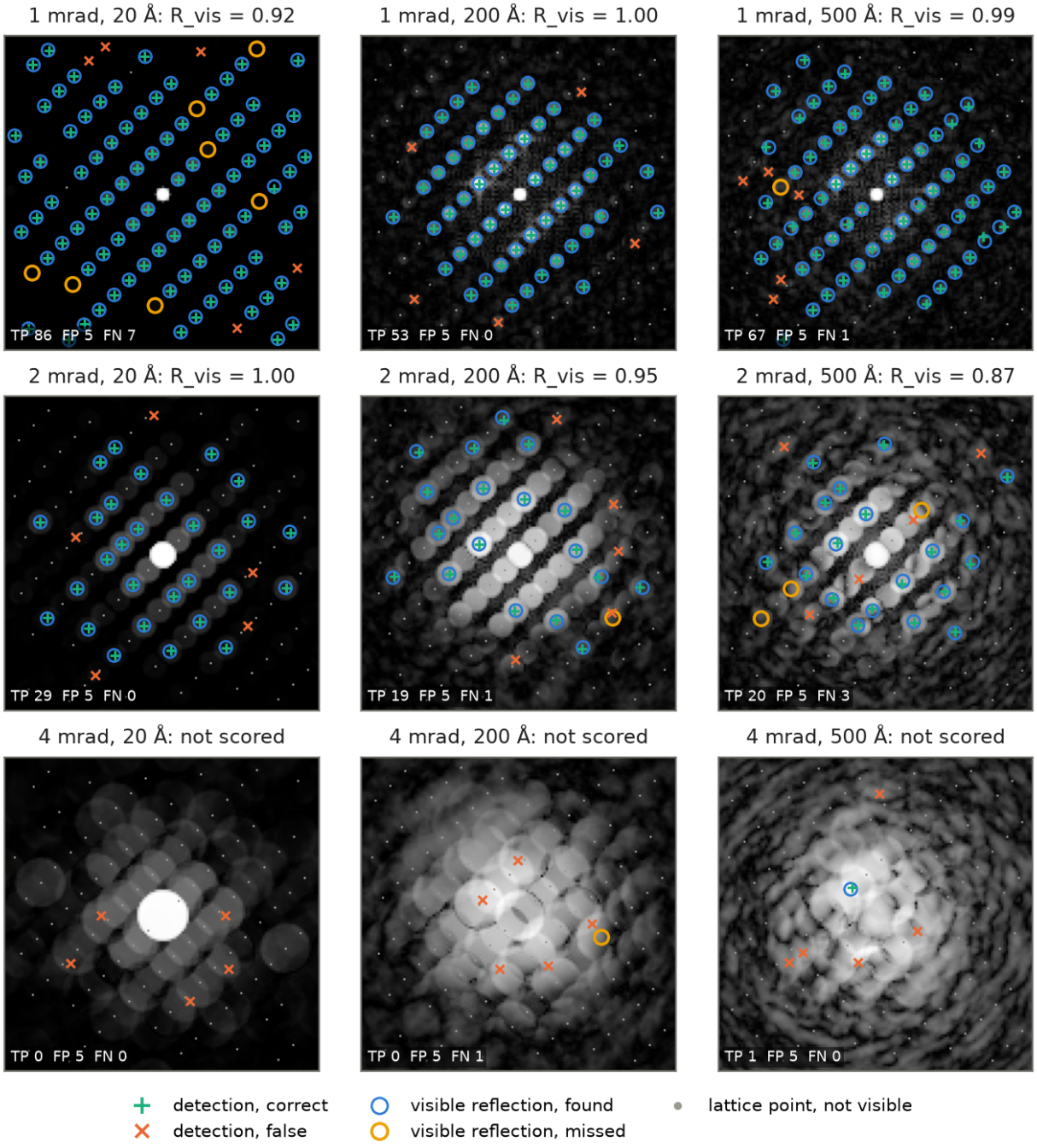

Figure 4: SiO_2 at 1, 2, 4 mrad (rows; $\eta = 0.60, 1.09, 2.29$) and 20, 200, 500 Å (columns). Each image uses its own threshold (at most five false detections). Green +: correct detection; orange ×: false detection; blue circle: visible reflection found; yellow circle: visible reflection missed; grey dot: lattice point that is not visible. At 4 mrad no reflection is visible and the detections fall between lattice points.

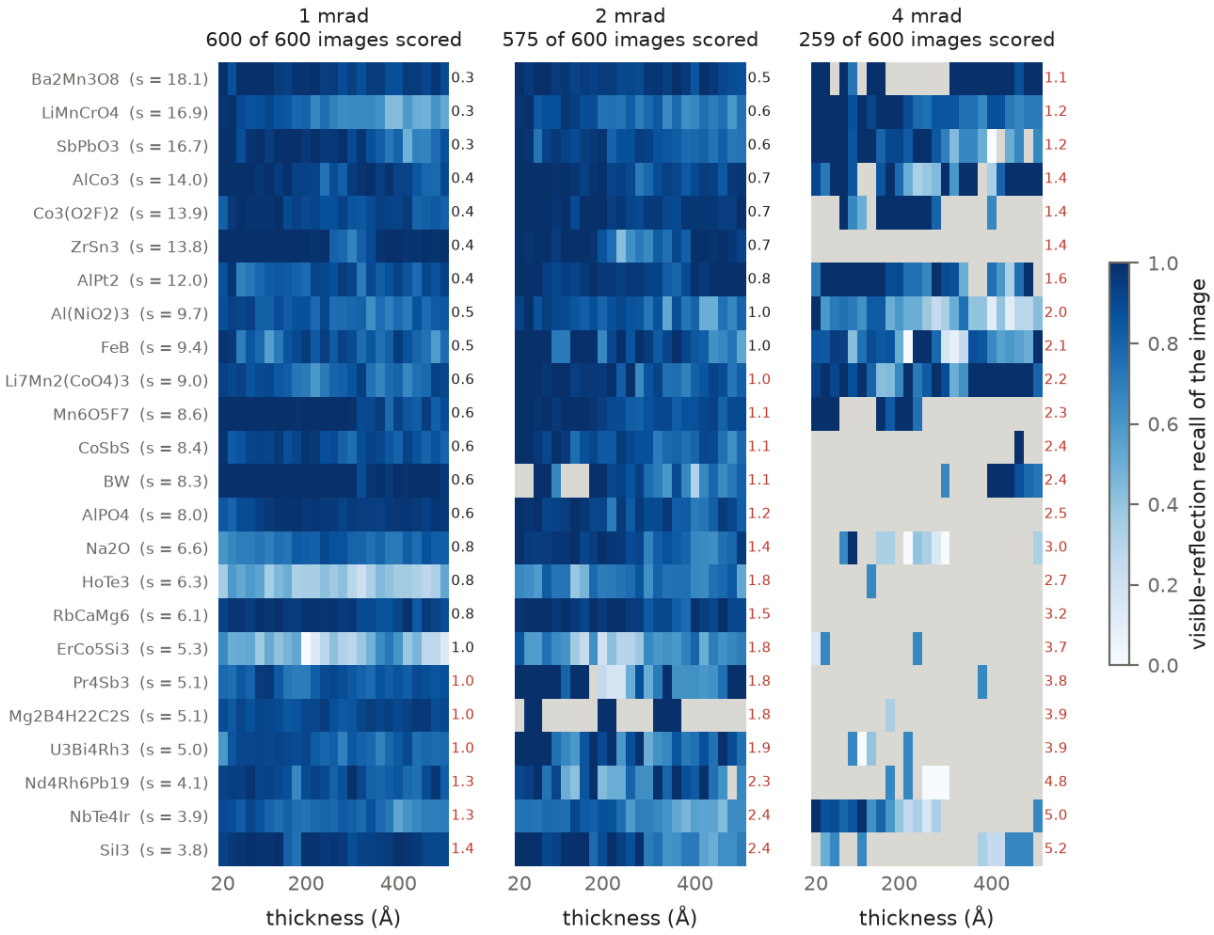

Figure 5: R_{vis} of every image of the 24 crystals (rows) at 25 thicknesses (columns) and three probe angles (panels). Rows are sorted by spot spacing s, densest lattice at the bottom; the number to the right of each row is η (red when $\eta \geq 1$). Grey: fewer than three visible reflections, not scored.

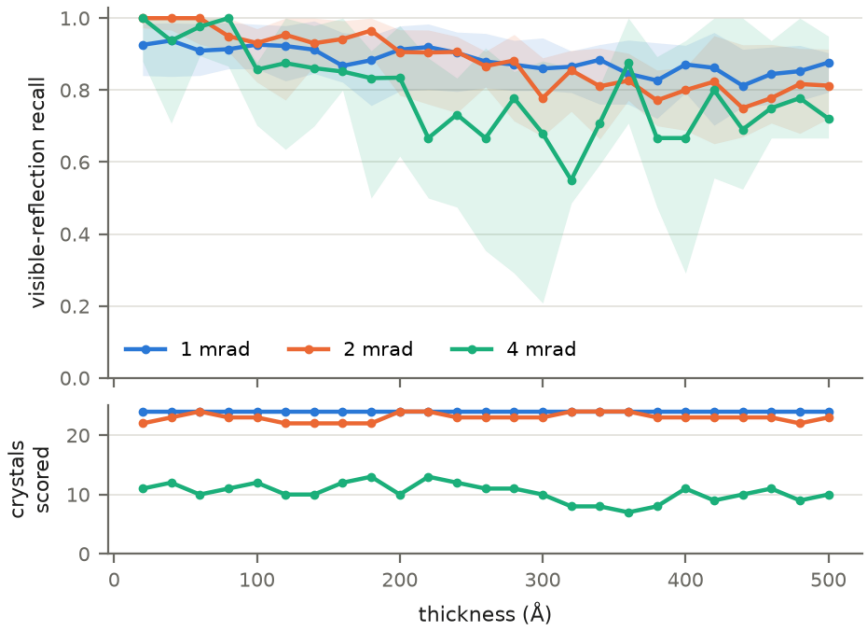

Figure 6: R_{vis} versus thickness for the 24 crystals: median (line) and middle 50 % (band) over the crystals, and the number of crystals scored at each thickness (bottom).

6 Why disk overlap makes locating disks difficult

6.1 When does the ring reach the neighbours?

The neighbouring disk begins at distance $s - r$ from a disk centre, and the filter ring ends at $r + g + w$ (Equation (4)). The ring stays on background as long as $r + g + w \le s - r$, that is

$$\eta \le \eta_c = 1 - \frac{g + w}{s}. \tag{9}$$

For $s = 14\,\mathrm{px}$, $\eta_c = 0.79$; for SiO_2 ($s = 8.6\,\mathrm{px}$), $\eta_c = 0.65$. Disks touch at $\eta = 1$. Because r grows with α and s is fixed, η doubles each time α doubles: over the 24 crystals the median η is 0.62, 1.13 and 2.37 at 1, 2 and 4 mrad, and 6, 15 and all 24 crystals have $\eta \ge 1$.

6.2 What the filter measures under overlap

For intuition, take linear intensities: a background b plus disks of height a_k centred at $\boldsymbol{c}_k$, $I(\boldsymbol{u}) = b + \sum_k a_k\,\mathbf{1}[\|\boldsymbol{u} - \boldsymbol{c}_k\| \le r]$. Inserting this into Equation (6) at the centre of disk 0 gives

$$S(\boldsymbol{c}_0) = a_0 + \sum_{k \ne 0} a_k \big[\varphi_D(d_k) - \varphi_R(d_k)\big], \tag{10}$$

where $d_k = \|\boldsymbol{c}_k - \boldsymbol{c}_0\|$ and $\varphi_D(d)$, $\varphi_R(d)$ are the fractions of the filter disk and of the filter ring covered by a disk of radius r at distance d. Three regimes follow:

1. $d_k \ge 2r + g + w$ for all neighbours ($\eta \le \eta_c$): $\varphi_D = \varphi_R = 0$ and $S(\boldsymbol{c}_0) = a_0$. The response is the disk's own contrast, independent of its neighbours.
2. $2r \le d_k < 2r + g + w$ ($\eta_c < \eta \le 1$): the disks do not yet touch, but $\varphi_R > 0$. Bright neighbours are *subtracted*; a weak disk next to strong ones can get a negative response.
3. $d_k < 2r$ ($\eta > 1$): the disks overlap and $\varphi_D > 0$ as well. Several neighbours cover the ring completely ($\varphi_R \to 1$), so the response at the centre becomes a difference of two sums of neighbour intensities with no particular sign. Meanwhile the lens-shaped regions where disks overlap form new bright patches of their own, which the filter can mistake for disks.

The true-log transform is monotonic, so the same regimes apply to T, although the terms no longer add exactly.

Figure 7a,b measures this on all 1 875 images. The fraction of the ring lying on neighbouring disks is zero up to $\eta \approx 0.8$ and reaches one beyond $\eta \approx 1.5$; of all the quantities measured it is the most closely tied to η (Spearman rank correlation 0.92). The median response at the true centres falls with it, from $z = 5.3$ for $\eta \le 0.5$ to 1.4 at $0.8 < \eta \le 1$ and 0.03 beyond $\eta = 2.5$ (Table 4): the true centres stop standing out from the rest of the pattern.

6.3 A controlled experiment

In real patterns several things change together with α. To isolate overlap, we generated synthetic patterns in which only the disk radius changes (Figure 8): a square lattice with $s = 14\,\mathrm{px}$ at a random rotation, flat disks with a one-pixel soft edge, reflection intensities drawn log-uniformly between 10^{-3} and 10^{-1}, a direct beam of intensity 1 and a weak diffuse background. The radius runs from 2 to 14 px ($\eta = 0.29$–2.0), with eight patterns per radius. The disks are combined in two ways: *incoherently*, $I = \sum_k I_k\,\mathbf{1}_{D_k}$, and *coherently* with random phases, $I = \big|\sum_k \sqrt{I_k}\,e^{i\phi_k}\,\mathbf{1}_{D_k}\big|^2$, which lets overlapping disks interfere. All disks count as reflections to be found.

The result separates the two causes. In both models every disk is found as long as $\eta \le 0.86$, a little beyond $\eta_c = 0.79$, while the response at the centres falls steadily from $z \approx 16$ to 3 as the disks grow. At $\eta = 1$ recall drops to 0.81, and beyond $\eta = 1.3$ the response at the centres is below

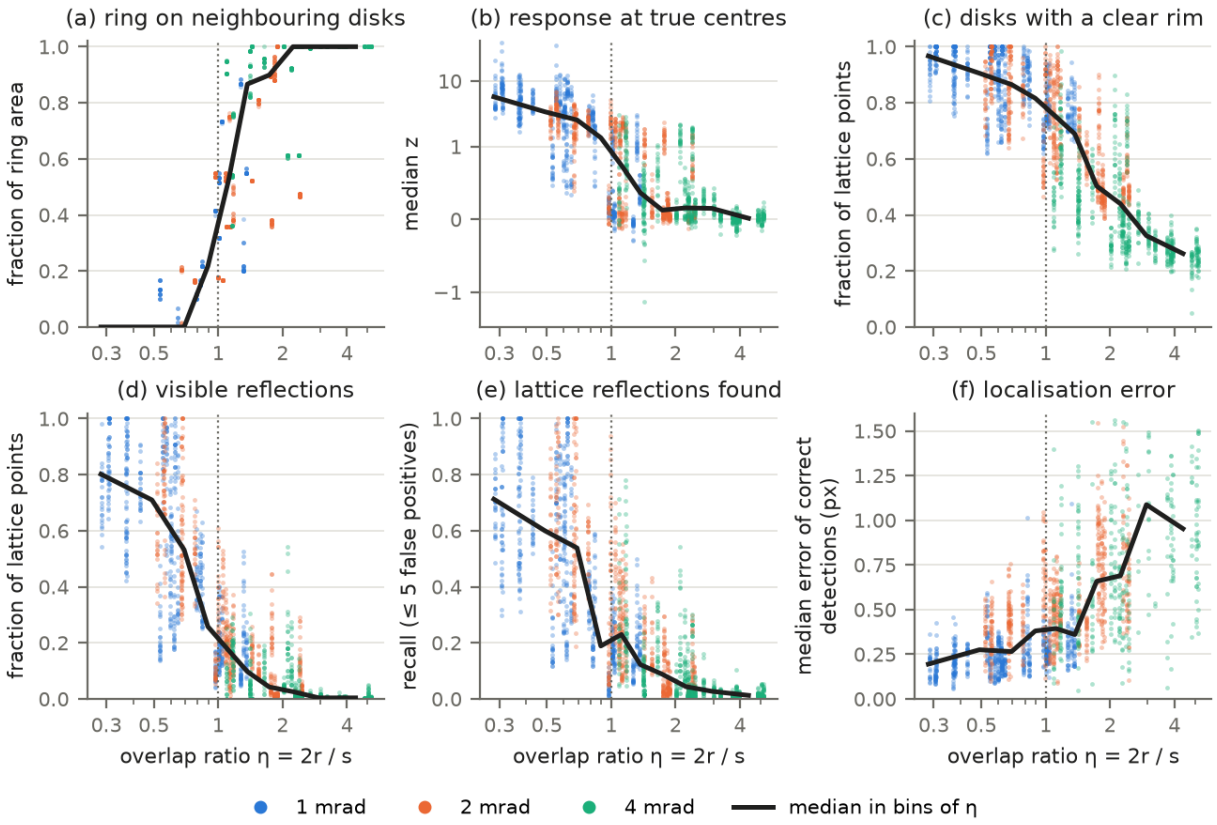

Figure 7: Every image of SiO_2 and the 24 crystals (dots, coloured by probe angle) versus its overlap ratio η; the black line is the median in bins of η; the dotted line marks $\eta = 1$. (a) Fraction of the filter ring lying on neighbouring disks. (b) Filter response at the true lattice points. (c) Fraction of lattice points whose disk still has a clear rim (intensity step of at least a factor 1.5). (d) Fraction of lattice points that are visible. (e) R_{all}. (f) Localisation error of correct detections.

0.6. With incoherent addition recall then settles between 0.38 and 0.54: geometry alone, the ring on the neighbours, is enough to lose half of the disks. With coherent addition it falls further, to 0.12–0.19: interference makes some overlapping disks darker than their surroundings, so even their own contrast a_0 in Equation (10) changes sign.

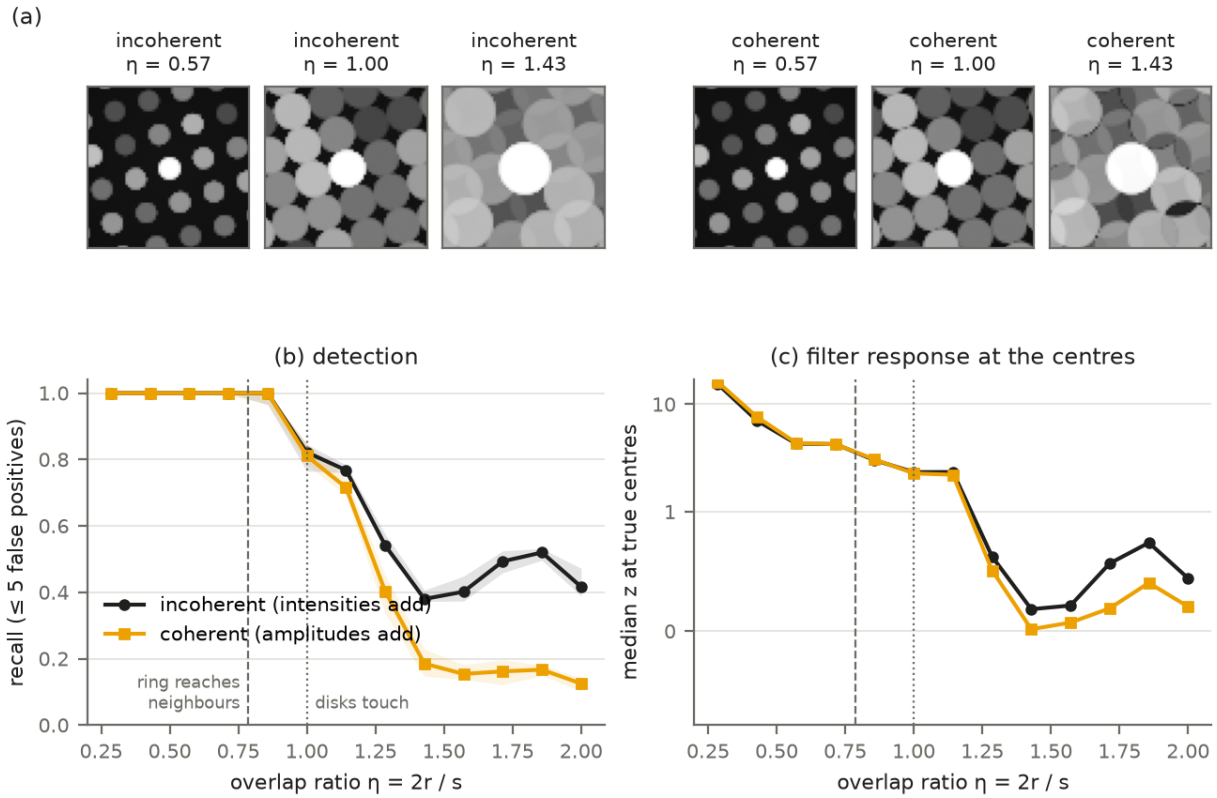

Figure 8: Synthetic patterns in which only the disk radius changes ($s = 14$ px). (a) True-log images at three overlap ratios, incoherent (left) and coherent (right) addition. (b) Recall at five false detections: median (line) and middle 50 % (band) of eight patterns. (c) Median response at the true centres. Dashed line: $\eta_c = 0.79$, Equation (9); dotted line: disks touch.

6.4 Disks lose their rim and their uniformity

The disk model behind the filter, a flat disk with a sharp rim, also fails in the real patterns. Figure 7c counts the lattice points whose disk still shows a clear step in intensity across its rim. This fraction falls from 0.95 for $\eta \leq 0.5$ to 0.48 at $1.5 < \eta \leq 2.5$ and 0.28 beyond. Among the disks that do keep a clear rim, the fraction that is *darker* inside than outside rises from 0.02 to 0.45 over the same range (Table 4), as in the coherent synthetic patterns. A dark disk has a negative response under Equation (6) and cannot be found by a filter that looks for bright disks.

6.5 Fewer visible reflections, and why R_{vis} hides the problem

The visibility test uses the same disk and ring as the filter: a reflection is visible if its disk is three times brighter than its ring. Under overlap the ring contains neighbouring disks, so the test fails for the same reason the filter does. The fraction of lattice points that are visible falls from 0.79 for $\eta \leq 0.5$ to 0.03 at $1.5 < \eta \leq 2.5$ (Figure 7d), and at 4 mrad the 24 crystals have a median of two visible reflections per image.

As a consequence R_{vis} is almost independent of η (Spearman -0.15): the reflections that are still visible are precisely those whose surroundings remain dark, and the filter finds them.

The loss appears instead in R_{all} (Spearman -0.84), which falls from 0.69 for $\eta \leq 0.5$ to 0.19 at $0.8 < \eta \leq 1.5$ and 0.02 beyond $\eta = 2.5$ (Figure 7e). Evaluating overlapped patterns with R_{vis} alone would therefore suggest that the detector still works, when in fact most of the lattice is no longer recovered.

6.6 Localisation becomes imprecise

Even when a disk is found, overlap shifts its peak. Neighbours enter the ring on one side more than the other, which pulls the maximum of S away from the centre, and a large disk with internal structure gives a broad, uneven plateau rather than a single sharp peak; with $r = 9.9$ px at 4 mrad, the peak window $\rho = 4$ px is also smaller than the disk, so one disk can produce several maxima. The median error of correct detections grows from 0.20 px for $\eta \leq 0.5$ to 0.38 px around $\eta = 1$ and 1.0 px beyond $\eta = 2.5$ (Figure 7f).

Table 4: All 1 875 images (SiO_2 and 24 crystals) grouped by overlap ratio: medians of per-image quantities. Clear rim: lattice points with an intensity step of at least $\times 1.5$ across the rim; dark: fraction of those that are darker inside. Scored: fraction of images with at least three visible reflections.

η	Images	Ring on neighb.	z at centres	Clear rim	Dark	Visible fraction	Scored	R_{all}	Error (px)
≤ 0.5	175	0.00	5.29	0.95	0.02	0.79	1.00	0.69	0.20
0.5–0.8	400	0.00	2.82	0.89	0.04	0.58	1.00	0.56	0.28
0.8–1.0	125	0.22	1.37	0.82	0.10	0.26	1.00	0.19	0.38
1.0–1.5	475	0.54	0.53	0.72	0.17	0.15	0.88	0.19	0.38
1.5–2.5	450	0.94	0.15	0.48	0.30	0.03	0.71	0.05	0.68
> 2.5	250	1.00	0.03	0.28	0.45	0.00	0.20	0.02	1.01

6.7 Worst case: $ZrSn_3$ at 4 mrad

$ZrSn_3$ shows all of these effects in one crystal (Table 2 and Figure 9). At 1 and 2 mrad ($\eta = 0.38$ and 0.68, both below $\eta_{\mathrm{c}} = 0.78$) it is one of the easiest crystals: $R_{\mathrm{vis}} = 1.00$ and 0.97, $R_{\mathrm{all}} = 0.94$ and 0.72. At 4 mrad the same lattice gives $\eta = 1.43$. The ring lies entirely on neighbouring disks, the response at the true centres falls from 5.8 to 0.25, only half of the lattice points keep a clear rim, and in 24 of the 25 images no reflection is visible. The detector finds 2 % of the lattice reflections; its strongest responses lie on the direct beam and on the lens-shaped overlap regions.

6.8 Summary

Overlap defeats the disk–ring filter through four effects, which set in at the same point, $\eta \approx 0.8$–1:

1. the ring falls on neighbouring disks and their intensity is subtracted from the response at the true centre (Equation (10)); this alone loses half of the disks in the synthetic test;
2. interference makes overlapping disks non-uniform and sometimes darker than their surroundings, which the filter cannot detect;
3. the visibility test fails for the same reason, so R_{vis} stays high while R_{all} collapses;
4. the peaks that remain are shifted and broadened, and the localisation error grows five-fold.

None of these is a defect of a parameter choice: they follow from looking for disks one at a time when the disks are no longer separate.

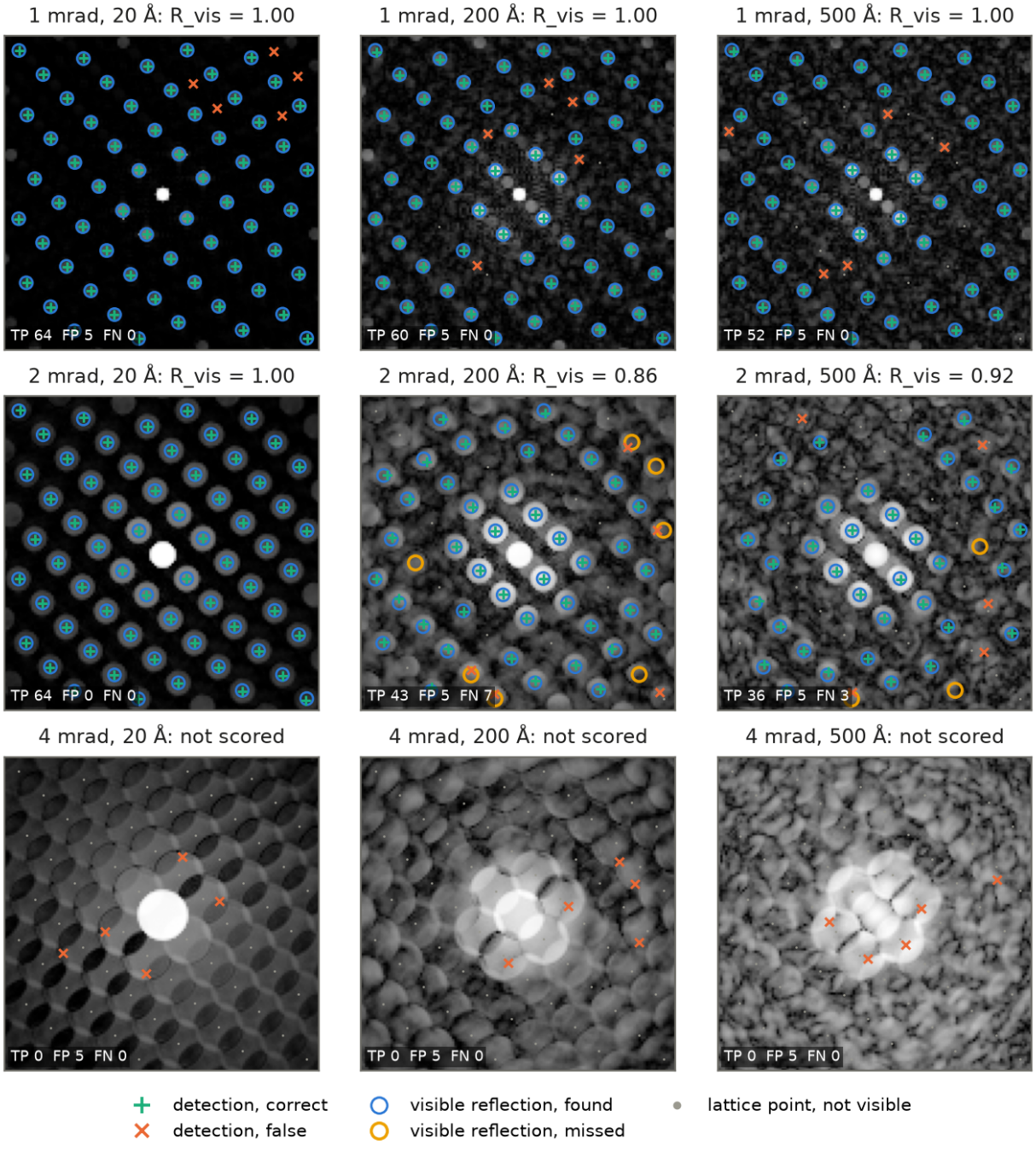

Figure 9: $ZrSn_3$, layout and symbols as in Figure 4 ($\eta = 0.38, 0.68, 1.43$). At 4 mrad the disks overlap everywhere, no reflection is visible, and the five false detections allowed per image fall on the direct beam and on overlap regions.

7 Disk–ring versus Sobel probe matching

Section 6 showed that under overlap the disk–ring filter fails because its ring no longer samples background and because some disks become darker than their surroundings. The rim of a disk, however, remains visible as an intensity step of either sign. This section tests a detector that matches only the rims: the gradient magnitude of the image correlated with the gradient magnitude of the probe.

7.1 Method

With G_x and G_y the horizontal and vertical Sobel kernels, the edge map of the true-log image is

$$E_T = \sqrt{(G_x * T)^2 + (G_y * T)^2}, \tag{11}$$

which is large at bright and at dark rims alike. The template E_P is the same quantity computed on the probe aperture P of the same probe angle, at the same pixel scale, after shifting P so that its centroid falls on the centre pixel of the template; it is cut to a half-width of $\lceil r \rceil + 3$ pixels and normalised to sum to one. The response is the cross-correlation

$$C(\boldsymbol{p}) = \sum_{\boldsymbol{u}} E_P(\boldsymbol{u})\, E_T(\boldsymbol{p} + \boldsymbol{u}), \tag{12}$$

computed by FFT with mirror padding, as for the disk–ring filter. Robust z-scoring, peak finding and sub-pixel refinement are unchanged (Equation (7)). On isolated synthetic disks made from the real probe, the detected centre is exact at integer positions and within 0.14 px at sub-pixel positions, the same as for the disk–ring filter.

7.2 Evaluation

Both detectors are run on all 1 875 images. Because the aim is to recover centres under overlap, every lattice reflection is a target, not only the visible ones. For each image the cutoff is the lowest z that keeps at most five false detections in that image, selected using the ground truth, as in Section 4. A detection is a *true positive* (TP) if an unmatched lattice point lies within 2 px, otherwise a *false positive* (FP; detections on the direct beam are ignored); a lattice point that no detection matches is a *false negative* (FN). The primary metric is $R_{\text{all}} = \text{TP}/(\text{TP} + \text{FN})$. Because the cutoff fixes the false detections at five per image, the comparison asks which detector finds more true centres at the same false-detection budget.

7.3 Results

Table 5 and Figure 10 summarise the comparison. At 1 mrad the two detectors are close (median R_{all} 0.41 for disk–ring, 0.37 for Sobel). At 2 mrad Sobel is much worse: it finds 10 465 true centres against 18 727 and loses in 514 of 625 images. At 4 mrad Sobel is better: it finds 7 387 true centres against 4 867, doubles the median R_{all} from 0.031 to 0.064, wins in 383 images and loses in 194, and its centres are more precise (median error 0.53 px against 0.89 px).

Sorted by overlap ratio (Figure 10b), the picture is consistent with Section 6. Sobel is better in 62 % of the images with $\eta > 2.5$, where many rims are dark and the disk–ring filter has nothing left to detect. It is worse in 88 % of the images with $0.5 < \eta \leq 0.8$ (median R_{all} 0.33 against 0.56), where the disks are still separate. The reason is that Equation (12) has no negative part: a position where the template circle crosses the rims of several neighbouring disks, or the very strong rim of the direct beam, scores as high as a true centre. In $ZrSn_3$ at 2 mrad and 200 Å, half of the 40 strongest Sobel peaks lie about one disk radius from a lattice point, against 7 % of the disk–ring peaks; in $AlCo_3$ at 2 mrad the Sobel false detections surround the direct beam (Figure 13). The disk–ring filter avoids this because its negative ring penalises positions next to bright structure.

Table 5: Disk–ring (DR) versus Sobel probe matching on SiO_2 and the 24 crystals (625 images per angle). All lattice reflections are targets; cutoff per image at five false detections, selected with the ground truth. TP, FN, FP: totals over the 625 images. Error: median over images of the median distance of the true positives to their lattice point.

α (mrad)	Detector	Median R_{all}	TP	FN	FP	Error (px)	Images better / worse
1	disk–ring	0.41	35 744	89 107	3 113	0.25	
	Sobel	0.37	31 412	93 439	2 889	0.24	150 / 394
2	disk–ring	0.24	18 727	106 501	3 095	0.55	
	Sobel	0.11	10 465	114 763	3 125	0.33	97 / 514
4	disk–ring	0.031	4 867	119 506	3 108	0.89	
	Sobel	0.064	7 387	116 986	3 124	0.53	383 / 194

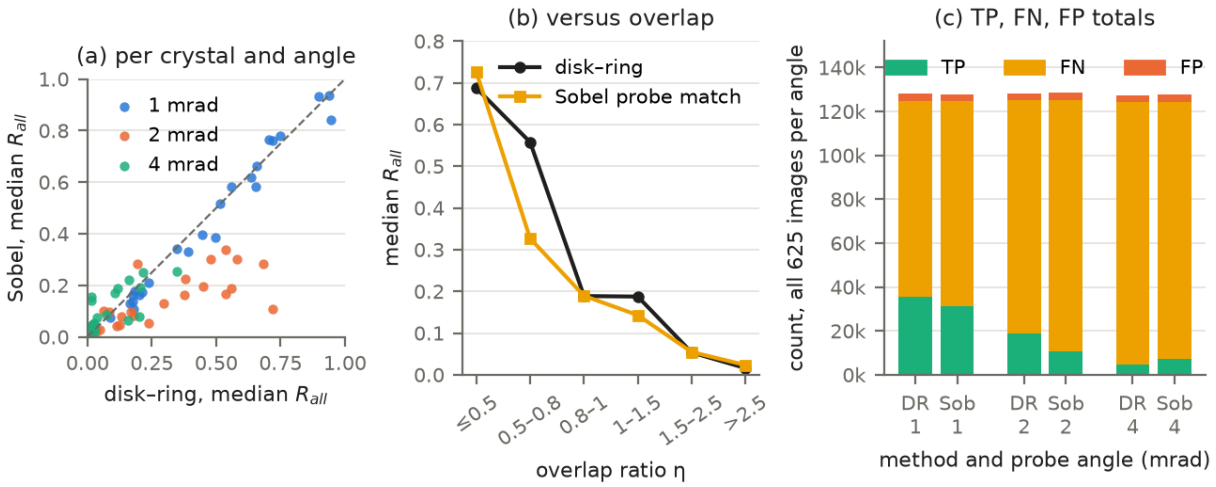

Figure 10: Disk–ring versus Sobel probe matching. (a) Median R_{all} over the 25 thicknesses, one point per crystal and probe angle; points above the diagonal favour Sobel. (b) Median R_{all} of all images in bins of the overlap ratio. (c) Total TP, FN and FP over the 625 images of each probe angle (DR: disk–ring, Sob: Sobel); TP + FN is the number of lattice reflections.

Even where it helps, Sobel does not solve overlap: at 4 mrad it still misses 94 % of the lattice reflections. The gain is largest in thin samples: at 4 mrad the mean gain in R_{all} is 0.06 up to 100 Å and vanishes beyond 400 Å. The best cases are $ZrSn_3$, $Co_3(O_2F)_2$ and CoSbS below 100 Å, where Sobel reaches $R_{\mathrm{all}} = 0.40$–0.59 against at most 0.11 for the disk–ring filter ($ZrSn_3$ at 20 Å: 33 of 64 centres against none). A practical combination is to use the disk–ring filter while $\eta \lesssim 1$ and the rim matcher beyond.

7.4 All materials

Figure 11 to Figure 35 show both detectors on every material at 200 Å, with the disk–ring filter in the top row and Sobel probe matching in the bottom row. Green +: true positive; yellow circle: false negative (lattice point missed); orange ×: false positive. The counts are given in each panel.

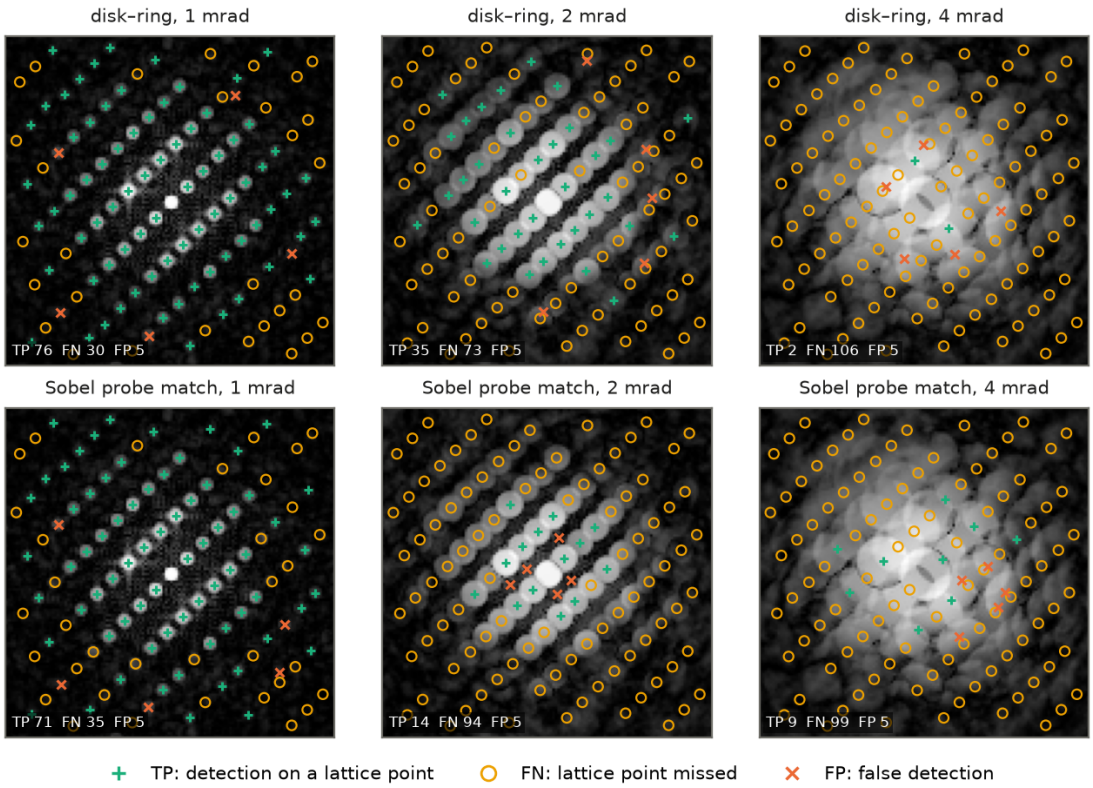

Figure 11: SiO_2, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

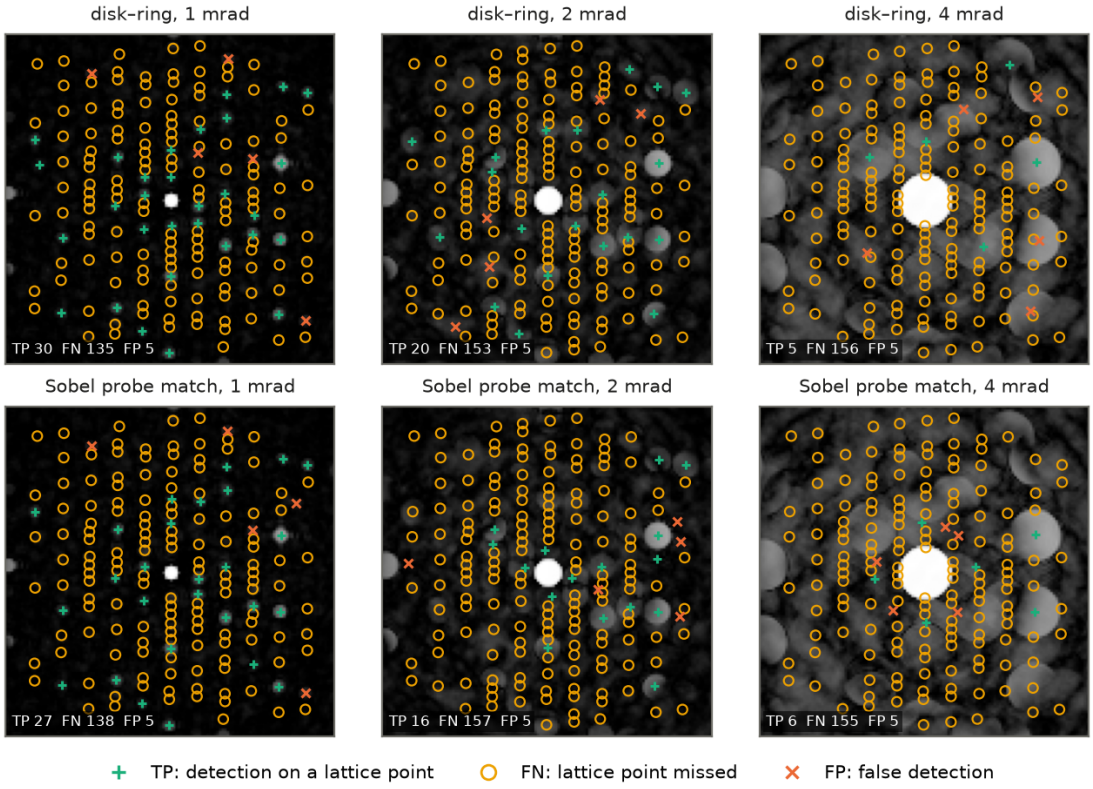

Figure 12: $Al(NiO_2)_3$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

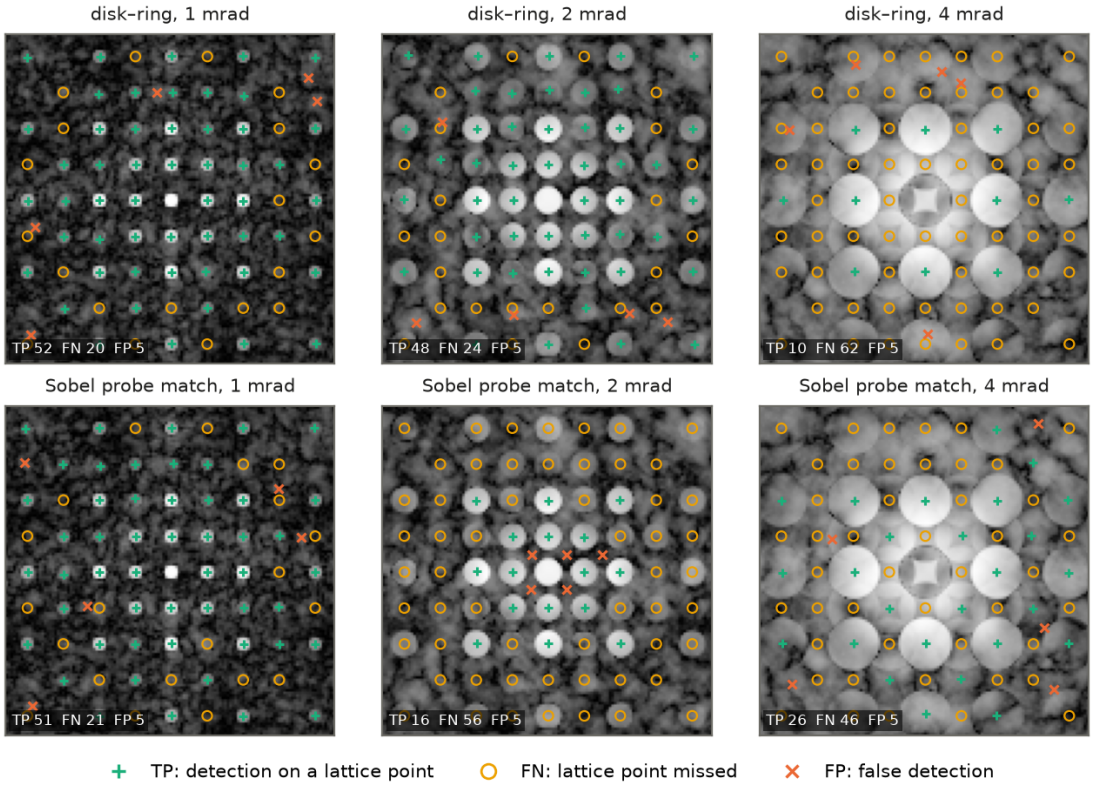

Figure 13: $AlCo_3$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

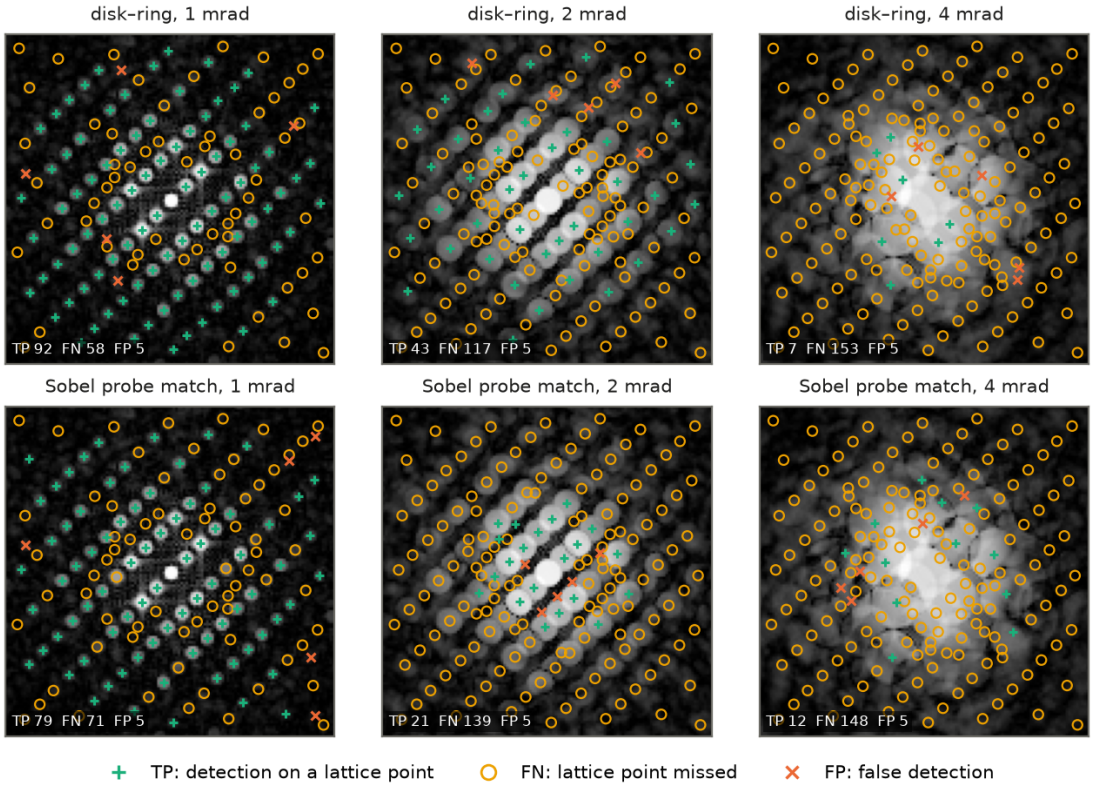

Figure 14: $AlPO_4$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

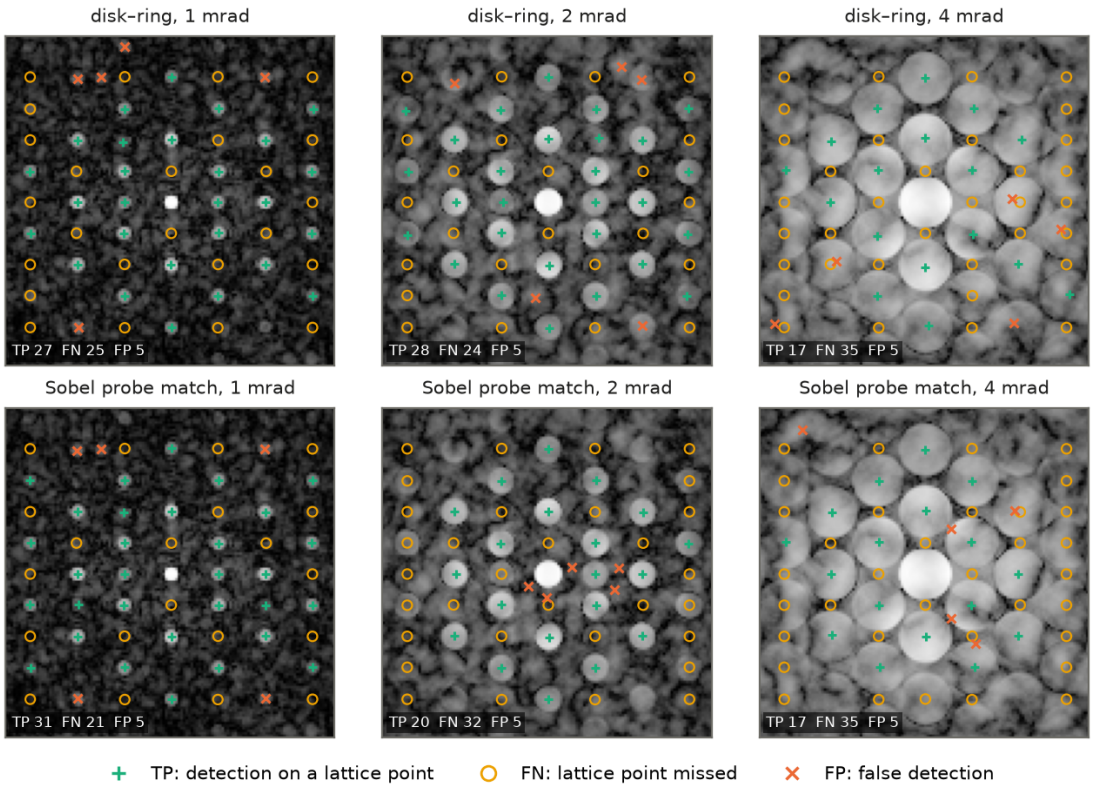

Figure 15: $AlPt_2$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

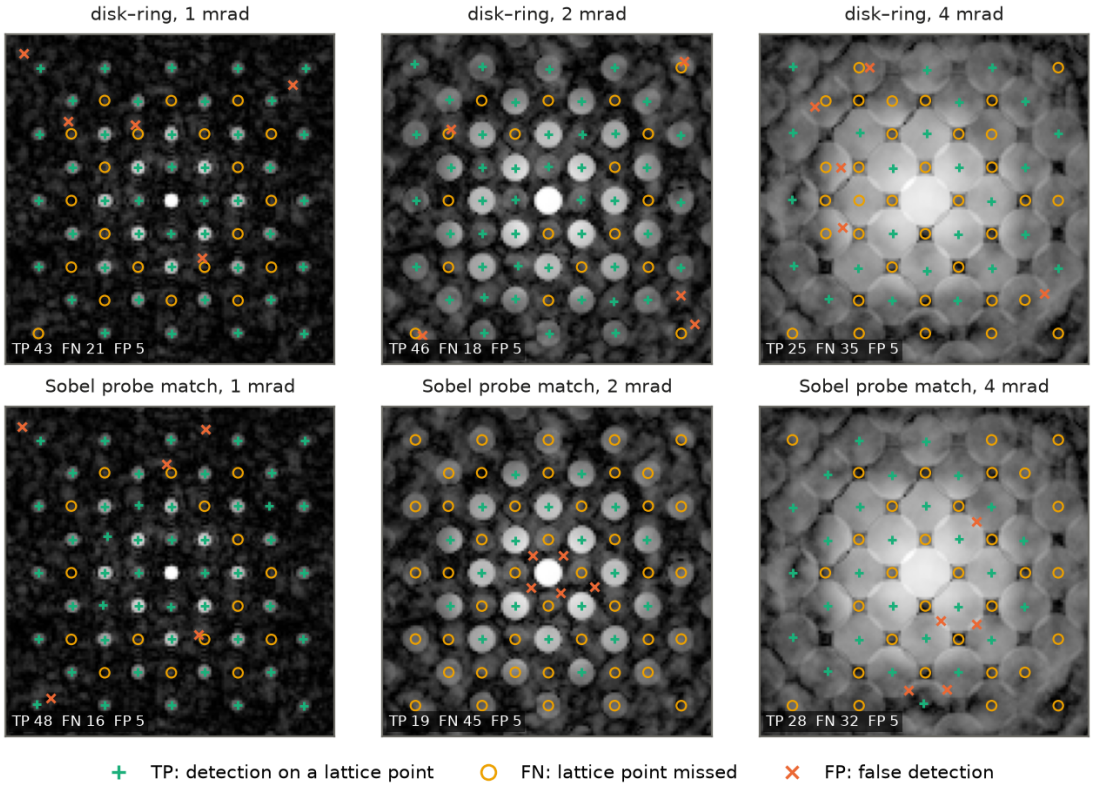

Figure 16: $Ba_2Mn_3O_8$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

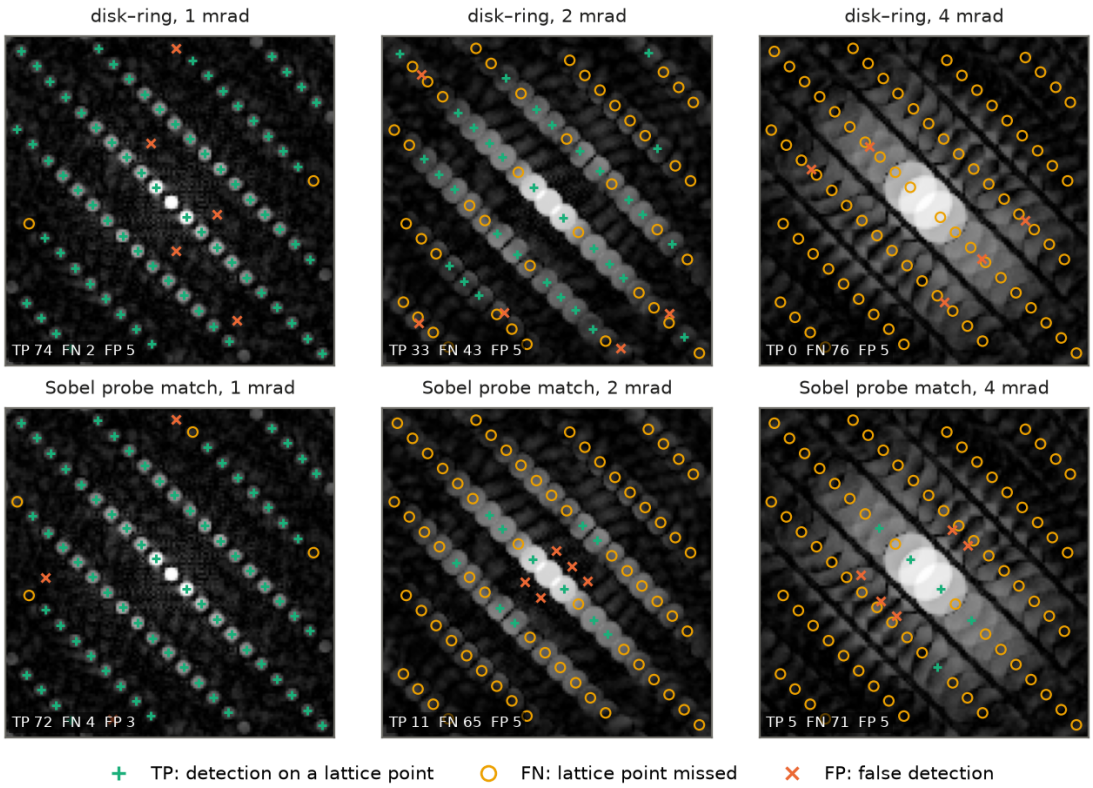

Figure 17: BW, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

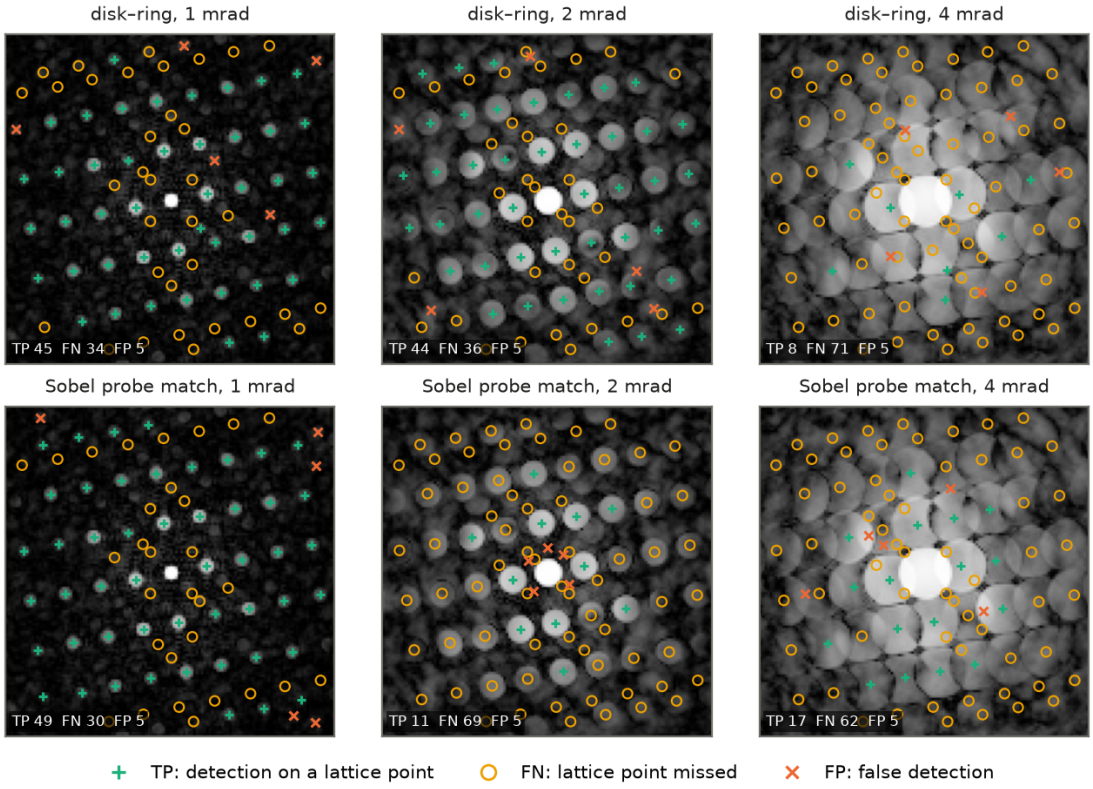

Figure 18: $Co_3(O_2F)_2$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

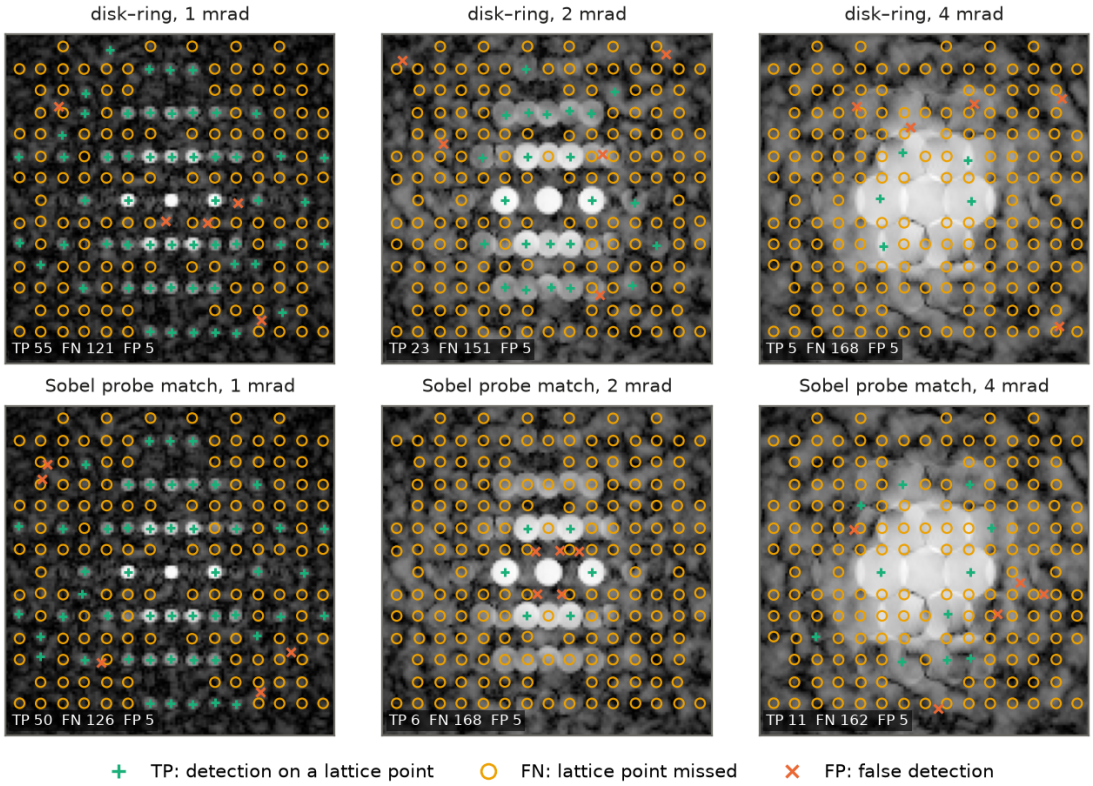

Figure 19: CoSbS, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

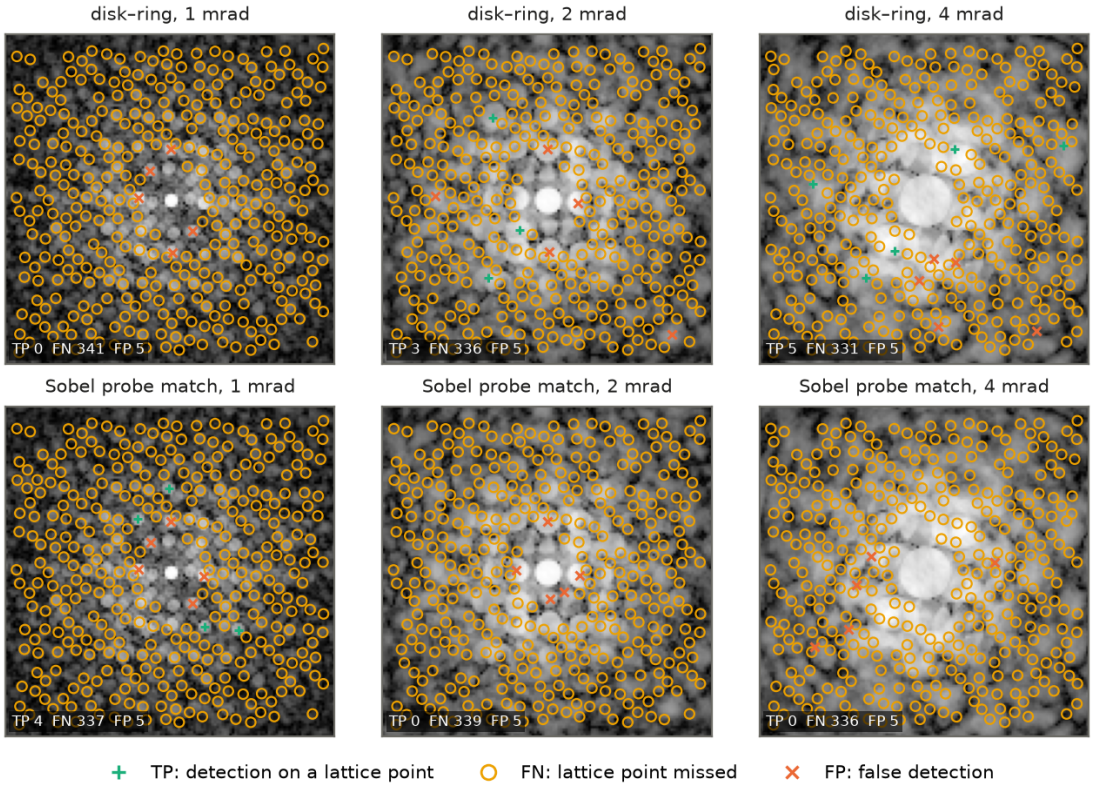

Figure 20: $ErCo_5Si_3$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

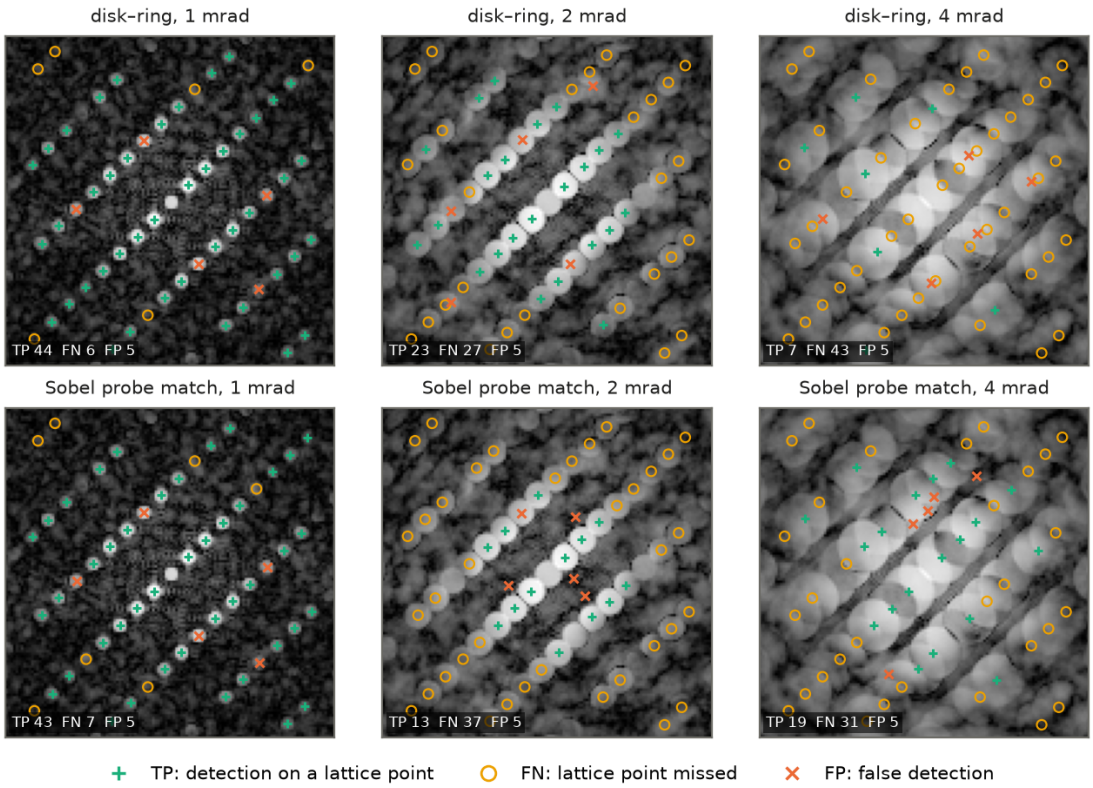

Figure 21: FeB, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

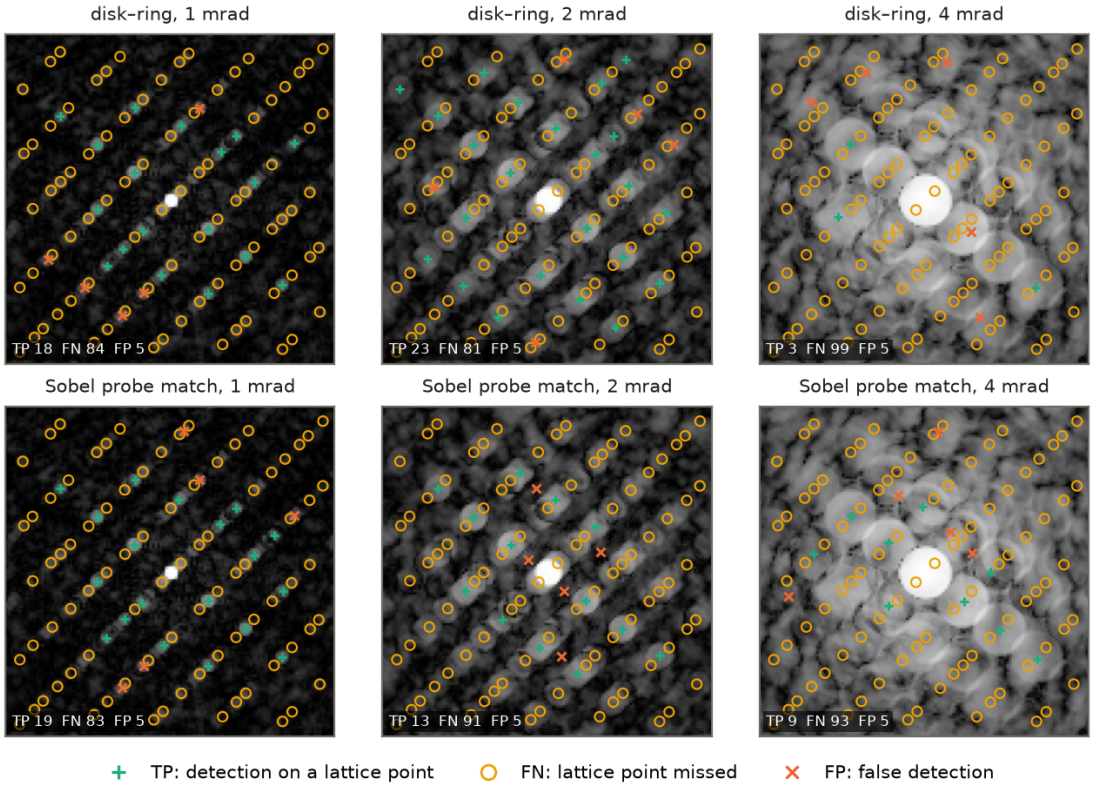

Figure 22: $HoTe_3$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

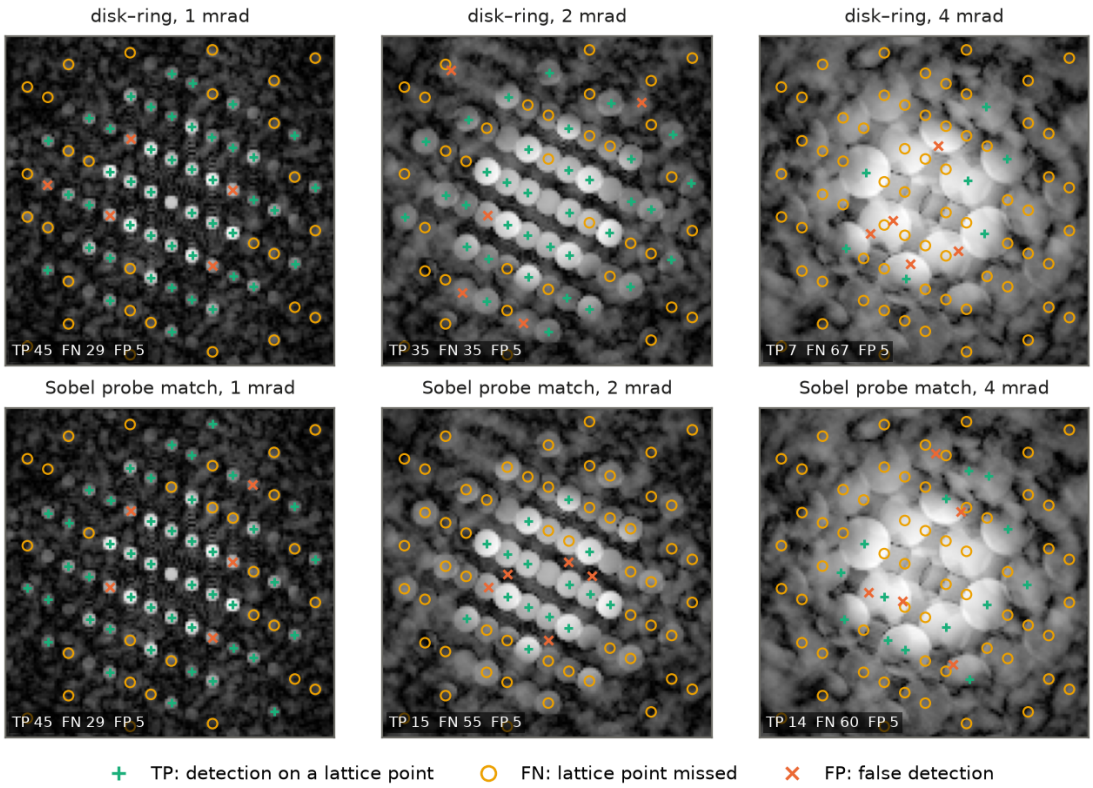

Figure 23: $Li_7Mn_2(CoO_4)_3$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

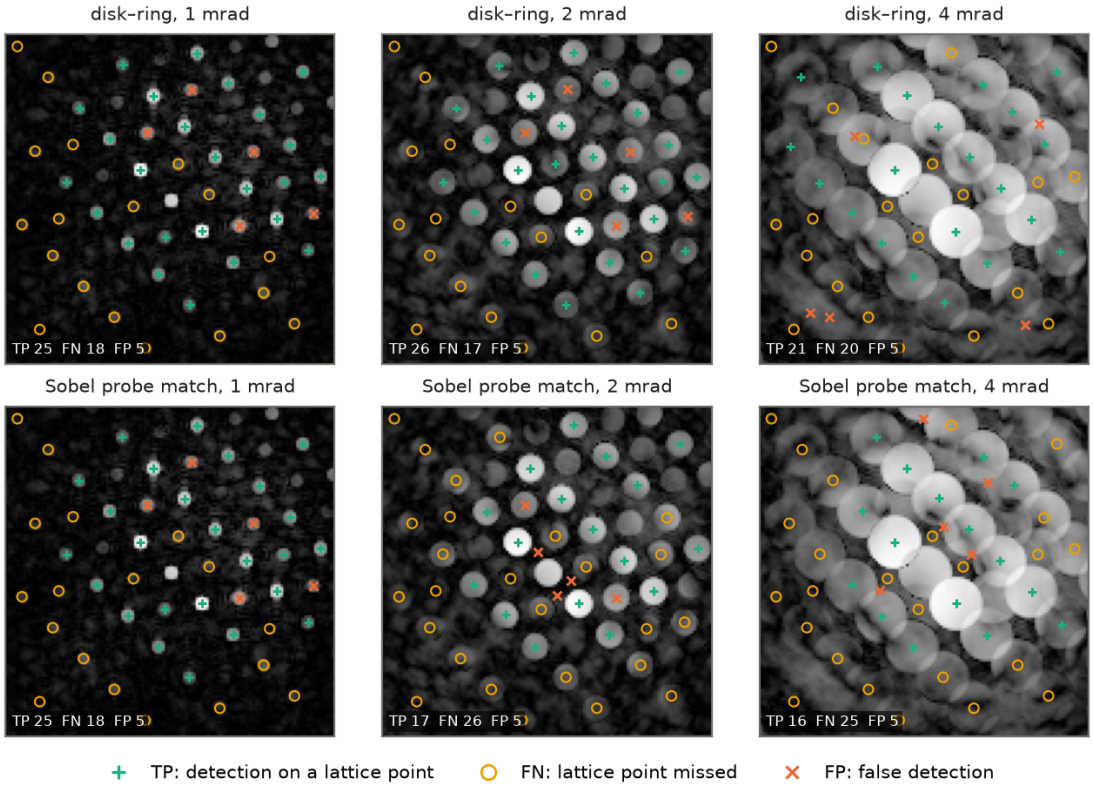

Figure 24: $LiMnCrO_4$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

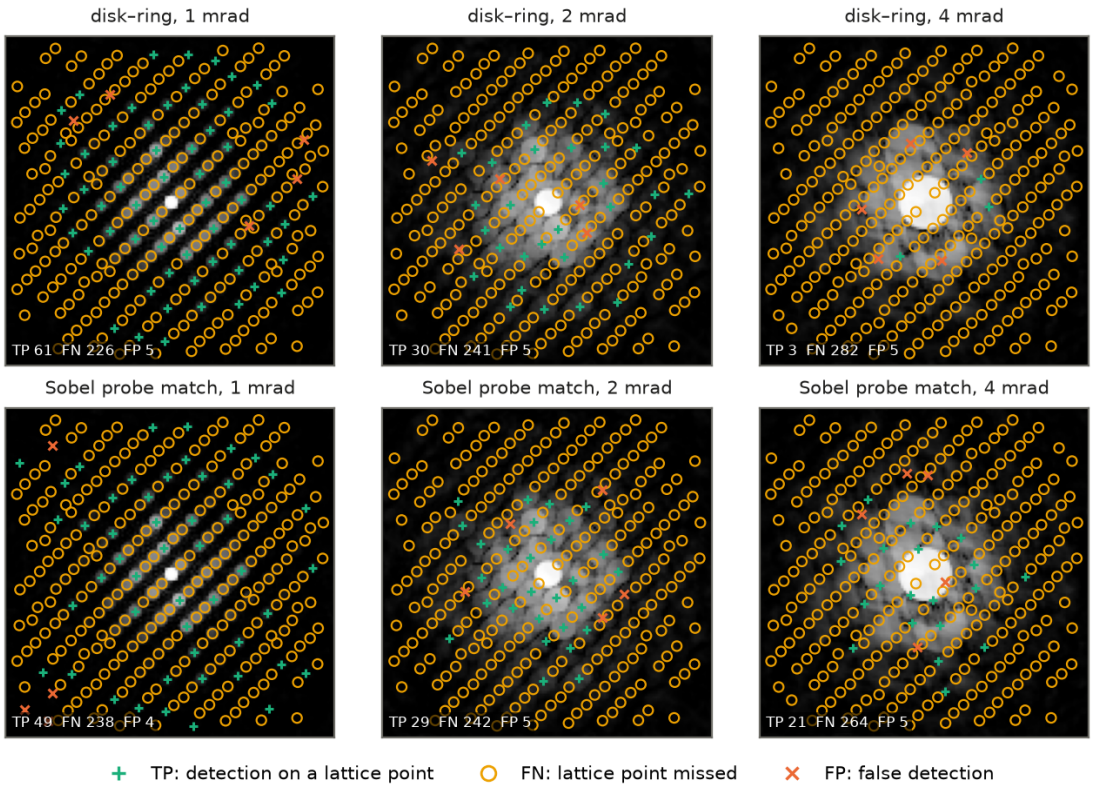

Figure 25: $Mg_2B_4H_{22}C_2S$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

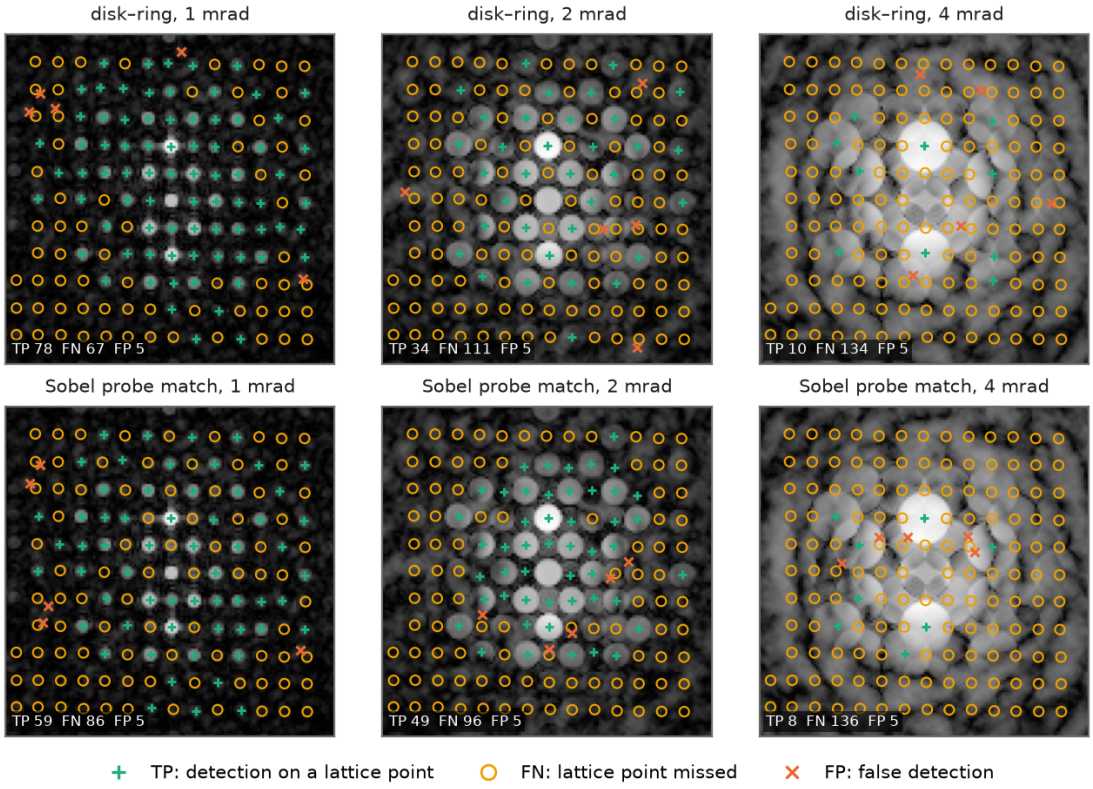

Figure 26: $Mn_6O_5F_7$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

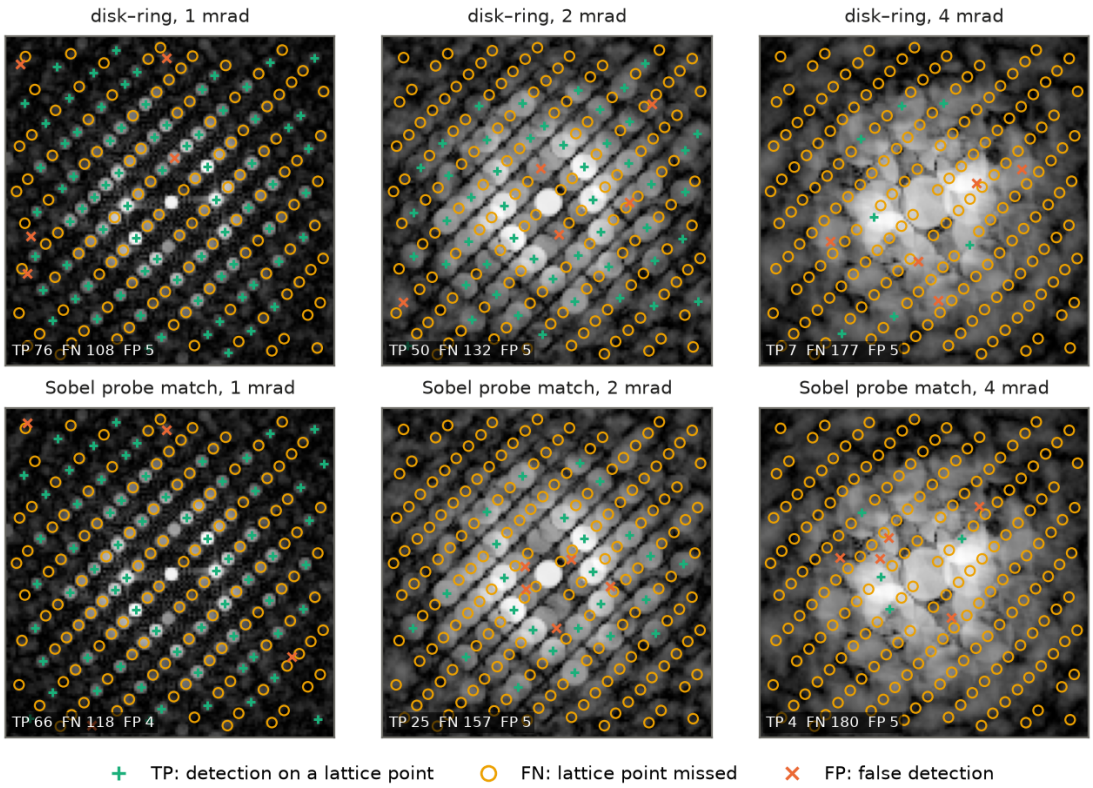

Figure 27: Na_2O, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

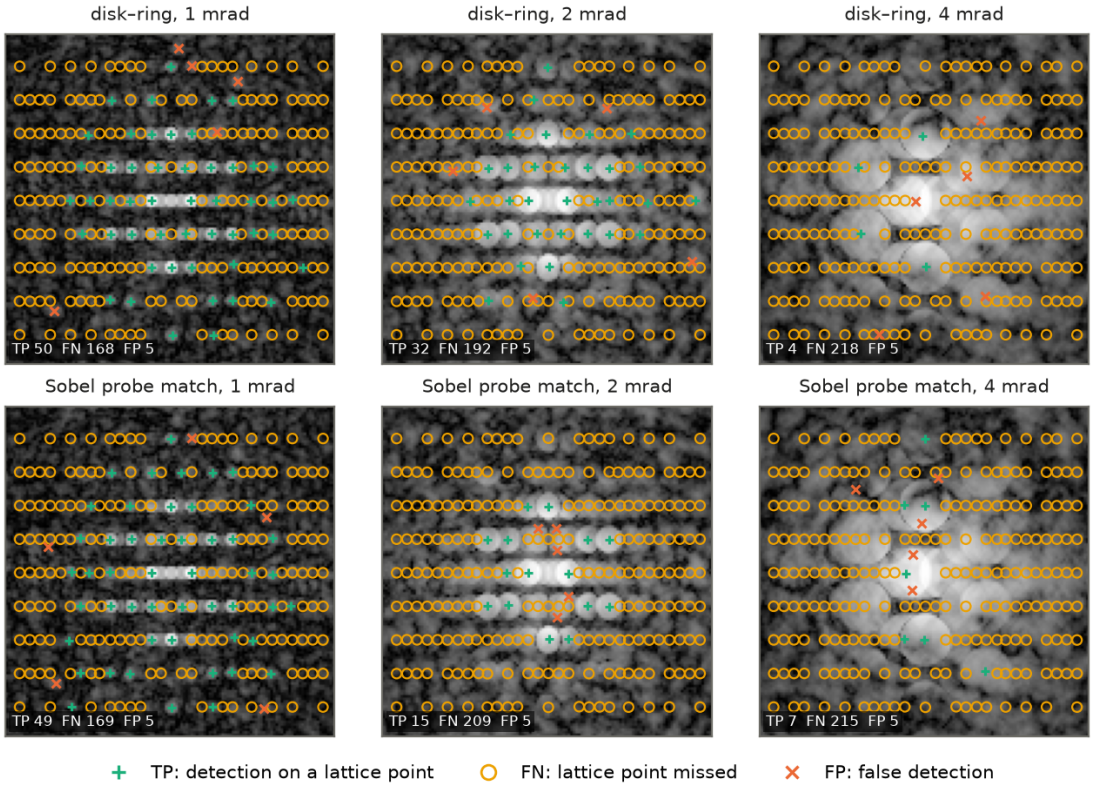

Figure 28: $NbTe_4Ir$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

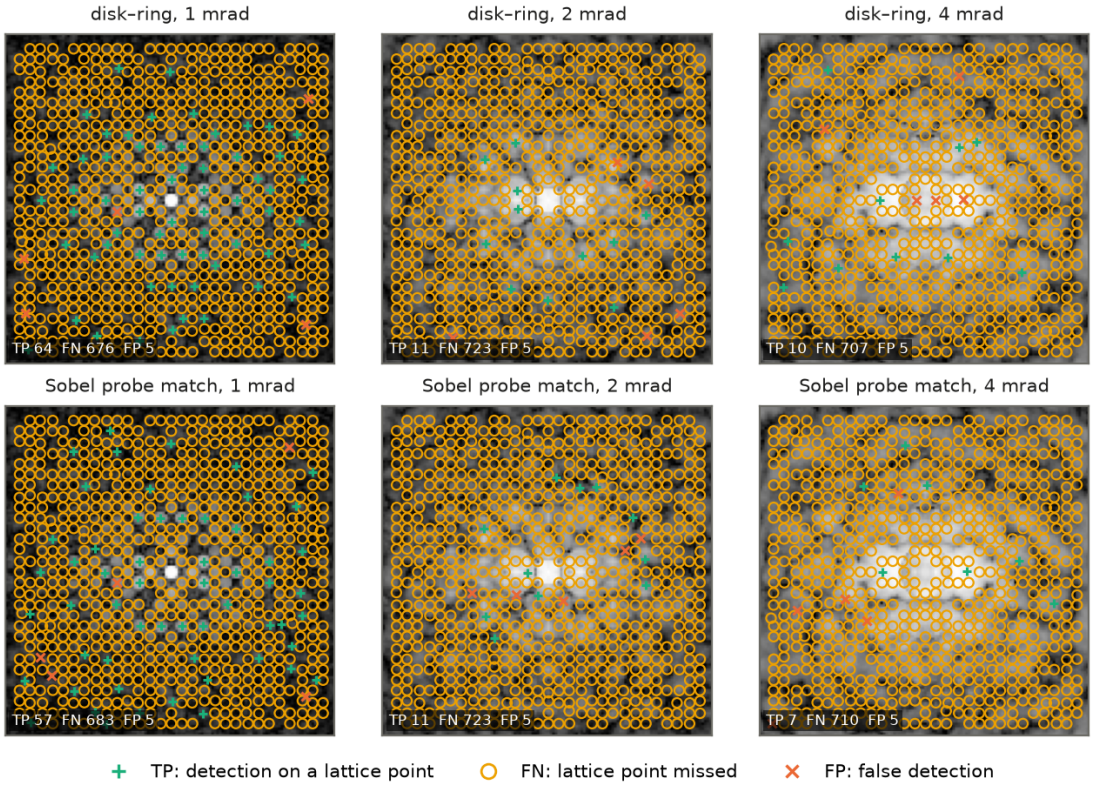

Figure 29: $Nd_4Rh_6Pb_{19}$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

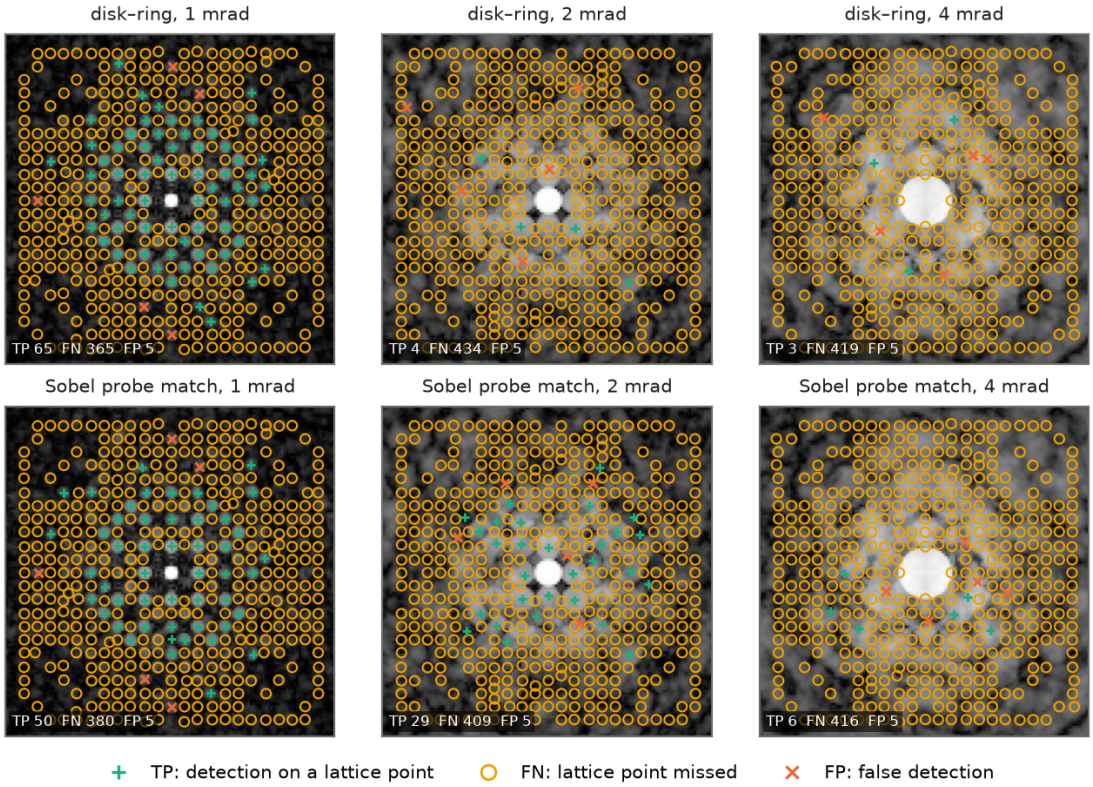

Figure 30: Pr_4Sb_3, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

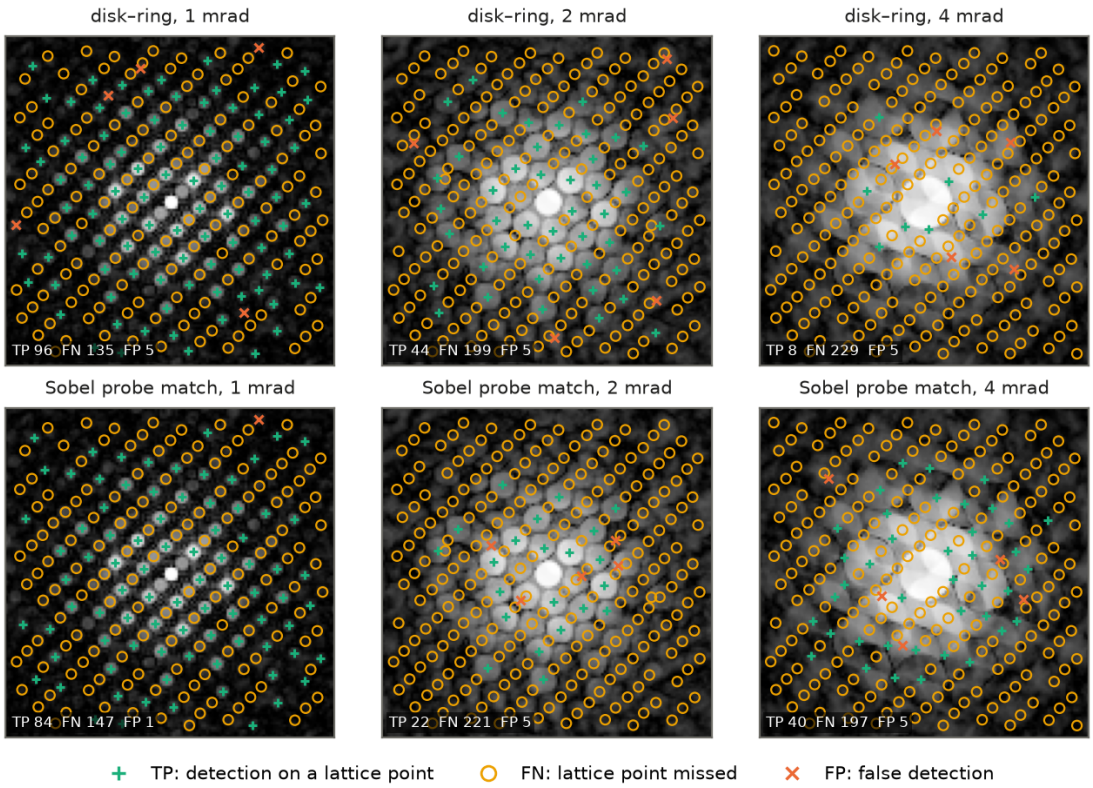

Figure 31: $RbCaMg_6$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

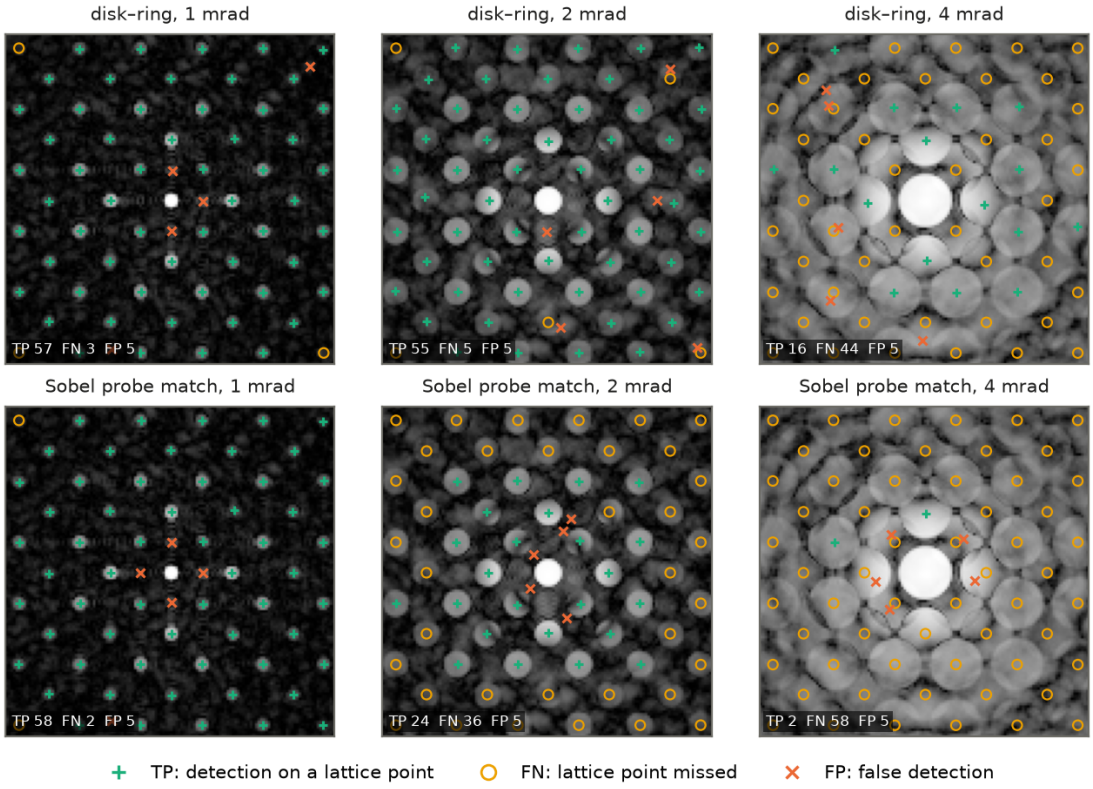

Figure 32: $SbPbO_3$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

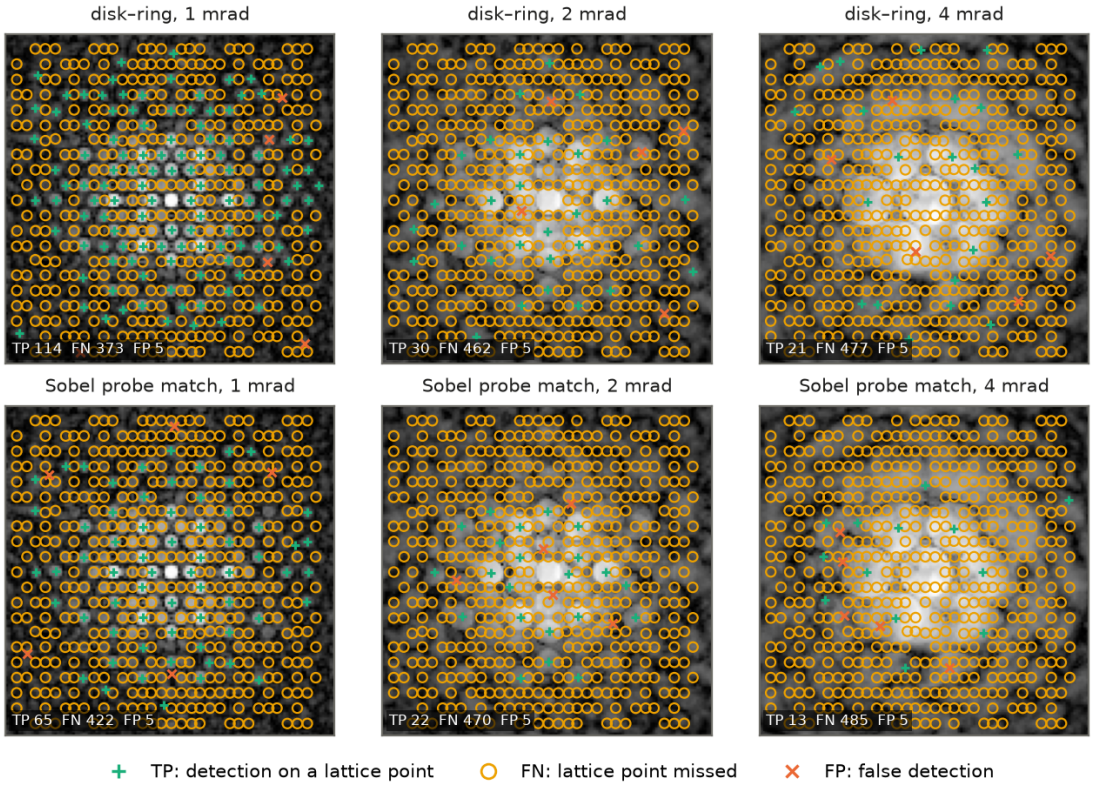

Figure 33: SiI_3, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

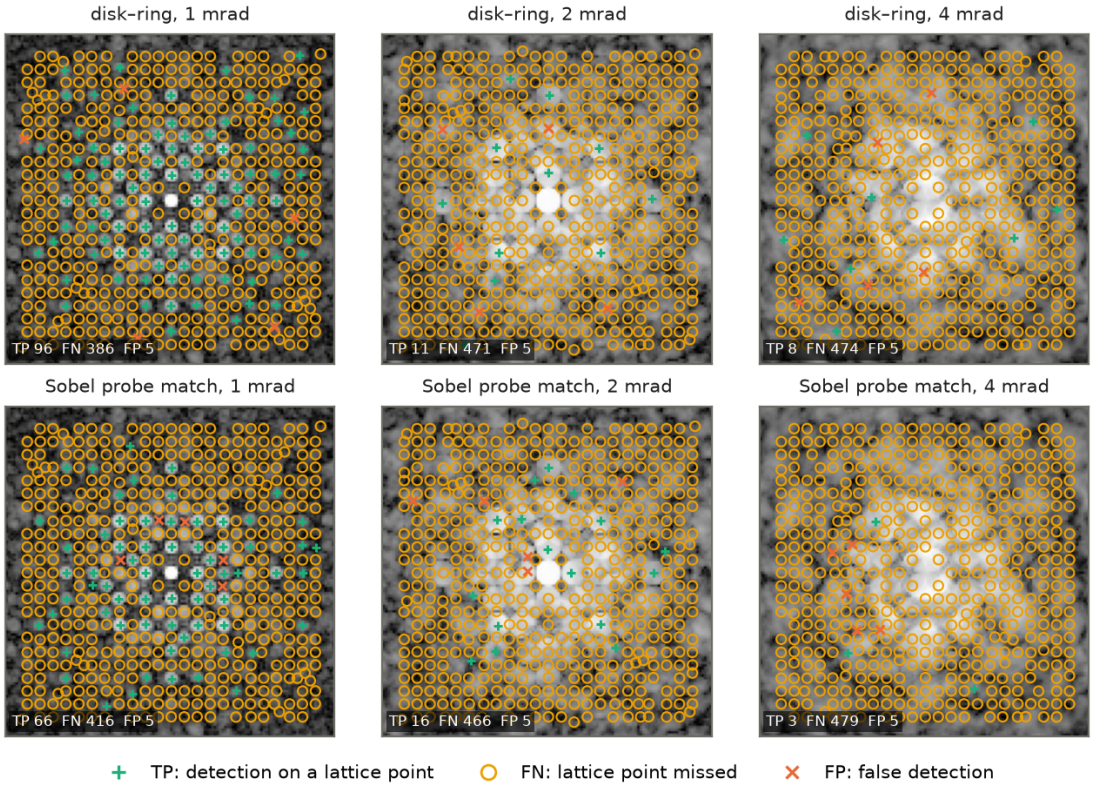

Figure 34: $U_3Bi_4Rh_3$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

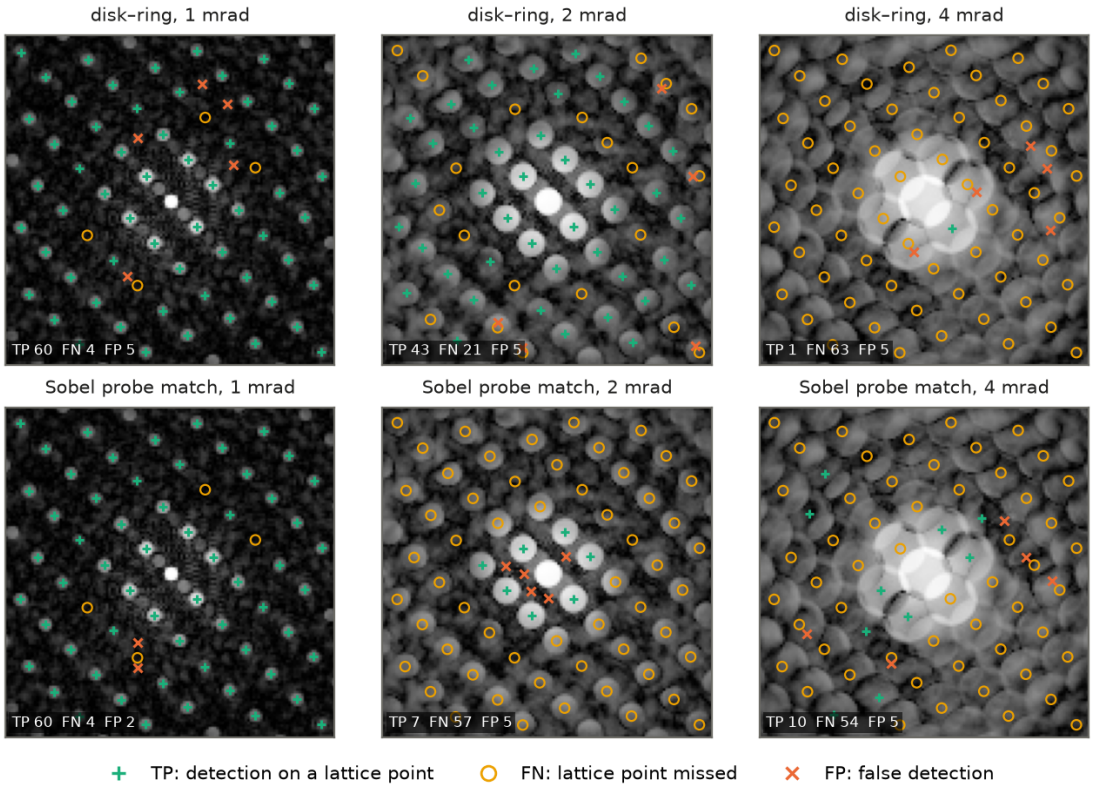

Figure 35: $ZrSn_3$, 200 Å: disk–ring (top) and Sobel probe matching (bottom) at 1, 2 and 4 mrad. Cutoff per image at five false detections, all lattice reflections as targets.

8 Practical guidance

Equation (9) with $r = \kappa\alpha$ gives the largest probe semi-angle at which the ring still lies on background:

$$\alpha_{\max} = \frac{s - g - w}{2\kappa} \approx \frac{s - 3\,\mathrm{px}}{5\,\mathrm{px/mrad}}. \tag{13}$$

Table 6 lists $\alpha_{\max}$ for representative lattices. It agrees with the results: $AlCo_3$ and $ZrSn_3$ ($\alpha_{\max} = 2.2$ mrad) work at 1 and 2 mrad and fail at 4 mrad; SiO_2 (1.1 mrad) works at 1 mrad and already loses half of its lattice reflections at 2 mrad. For dense lattices ($s < 5$ px) no practical probe angle keeps the disks apart. When a larger angle is required, rim matching (Section 7) recovers some centres, but the disks essentially have to be found jointly rather than one at a time, for example by fitting overlapping disks constrained to a lattice to the whole pattern.

Table 6: Largest probe semi-angle that keeps the filter ring on background, Equation (13), with $\kappa = 2.46$ px/mrad (mean of r/α over the three angles).

Crystal	s (px)	$\alpha_{\max}$ (mrad)	Observed
$Ba_2Mn_3O_8$	18.1	3.07	good at 1 and 2 mrad
$AlCo_3$, $ZrSn_3$	13.8–14.0	2.2	good at 1 and 2 mrad, fails at 4 mrad
SiO_2	8.6	1.14	good at 1 mrad, R_{all} halved at 2 mrad
$HoTe_3$	6.3	0.67	$R_{\mathrm{vis}} = 0.36$ already at 1 mrad
SiI_3	3.8	0.16	$\eta = 1.4$ already at 1 mrad

9 Limitations

- The threshold of each image is chosen on that image, which is optimistic; a fair protocol fixes it on some images and applies it to others.
- The spacing s counts every lattice point with at least 5 % of the strongest V, including reflections that are weak in M. For some dense lattices η therefore overstates the overlap between the bright disks (six crystals have $\eta \geq 1$ at 1 mrad but still $R_{\mathrm{vis}} \geq 0.8$).
- The visibility test shares the filter's disk–ring geometry, so part of the fall in visible reflections under overlap is a property of the definition; R_{all} does not have this problem.
- The patterns are simulated and noise-free. Detector data add Poisson noise, which would require a variance-stabilising transform before the logarithm.

A Parameters and code

Table 7: All parameters. Only ε and ρ were chosen from the data; r follows from the probe.

Parameter	Value	Role
ε	10^{-4}	true-log floor, Equation (2)
r	2.6 / 4.7 / 9.9 px	disk radius at 1 / 2 / 4 mrad, Equation (3)
g, w	1 px, 2 px	gap and ring width, Equation (4)
ρ	4 px	minimum distance between detected peaks
Matching radius	2 px	detection counted as correct
Visibility	disk $\geq 3\times$ ring, $\geq 10^{-3}$ of brightest	visible reflections
Operating point	≤ 5 false detections per image	threshold, chosen per image

Code and data.

- Method: `notebooks/dummy_data_explore.ipynb`, functions `true_log01`, `disc_ring_kernel`, `disc_centre_map` and `pick_peaks`.
- Per-image evaluation: `notebooks/scripts/run_per_image.py`; results in `notebooks/results/per_image_recall.csv`.
- Overlap analysis and synthetic experiment: `notebooks/scripts/run_overlap_analysis.py`; results in `notebooks/results/overlap_real.csv` and `overlap_synthetic.csv`.
- Sobel probe matching: `sobel_probe_map` in the notebook; `run_per_image.py` (method "true log + Sobel probe match"), results in `notebooks/results/per_image_sobel_all.csv`.
- Figures: `notebooks/tutorial/make_figs.py` and `make_figs_sobel.py`.